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Course: SmartGrids
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SmartGrid Project

The Government of Catalunya has commissioned a comprehensive assessment of a 110 kV transmission network supplying the cities of Lleida, Tàrrega, Manresa and Montblanc, with the objective of ensuring secure, efficient and sustainable operation under evolving demand and decarbonisation constraints. The existing system is characterised by a single large generation source at the Vandellòs nuclear power plant and a strong dependence on power exchange through the Terrassa interconnection, which together raise concerns regarding demand coverage, voltage performance, reliability and operating costs.

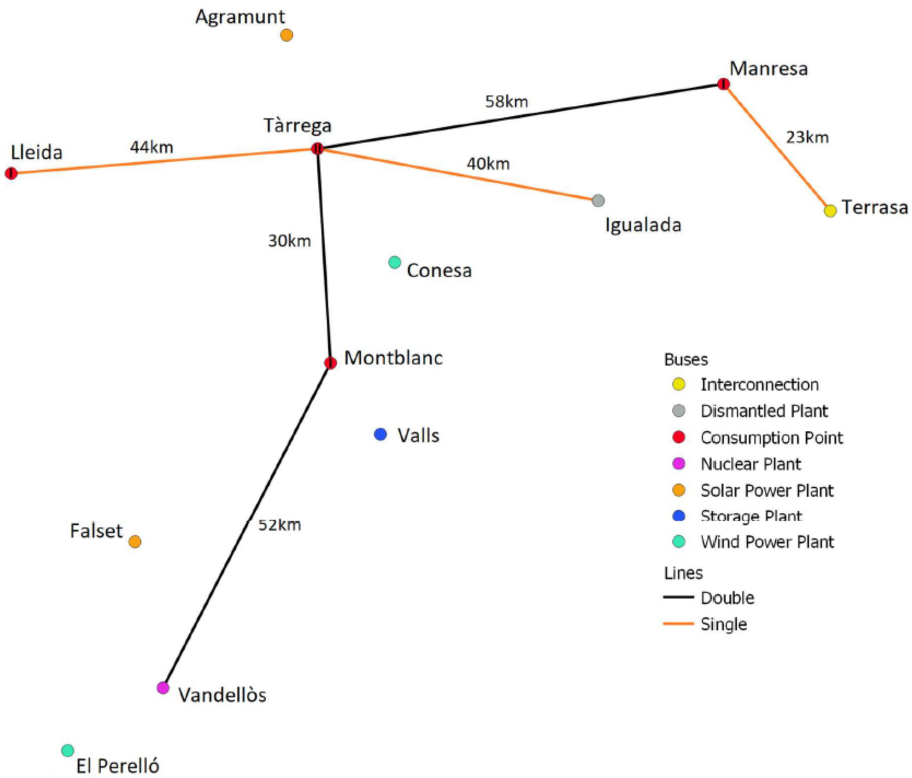

Figure 1: 110 kV transmission network topology under study, serving Lleida, Tàrrega, Manresa and Montblanc.

This report addresses these challenges by developing a detailed steady-state and time-series model of the network in PandaPower, based on real 24-hour demand and generation profiles obtained from the Spanish TSO (REE) for a representative working day in 2024. The model is used to analyse demand coverage, voltage deviations, equipment loading, losses and failure-related costs, thereby identifying the main operational bottlenecks and structural weaknesses of the current grid.

Building on this baseline analysis, the study is structured in two main planning phases. Phase 1 focuses on upgrading the existing network by adding new 110 kV transmission lines to satisfy the N-1 criterion, eliminate voltage violations and respect loading and loss constraints, while estimating both the required investment and the resulting annual operating costs. Phase 2 evaluates three alternative development strategies: integration of new solar and wind generation plants, deployment of a large-scale energy storage system, and installation of dynamic reactive power compensation. For each solution, the technical performance and economic impact are assessed through 24-hour load flow simulations, contingency analysis and cost estimation, leading to a final comparison and recommendation of the most cost-effective option to enhance the robustness and sustainability of the Catalan transmission network.

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1 Study of the current system

This section summarises the main technical, electrical and economic parameters used to model the current operation of the 110 kV transmission network. The data presented in the following tables constitute the input information for the steady-state analysis, reliability assessment and planning studies carried out throughout this work.

Specifically, the tables include the nominal generation and load levels, the electrical characteristics of the transmission lines and conductors, the main parameters of the HV/MV transformers supplying the distribution networks, as well as the economic and reliability assumptions adopted for the evaluation of different operating and planning scenarios. Finally, the renewable generation projects considered as potential future developments are also described.

Table 1: Generation and load data of the 110 kV system

Element	**Value**	**Connection level**
Nominal power of nuclear plant (Vandellòs)	215 MW	HV (110 kV)
Peak power consumption Type I load	65 MW	MV (20 kV)
Peak power consumption Type II load	115 MW	MV (20 kV)

Table 2: Transmission line and conductor characteristics

Parameter	**Value**
Conductor type	ACSR Cardenal
Composition	54 Al + 7 Ac
Diameter	30.42 mm
Resistance	0.062 Ω/km
Weight coefficient k_g	0.809
Maximum current	888.98 A
Simple line geometry	$a = 10$ m, $b = 4$ m
Double line geometry	$a = c = 7$ m, $b = 8$ m, $d = e = 7.5$ m
Quarter of the year analysed	T3 (July–September)

Table 3: HV/MV transformer characteristics

Rated Power	**Voltage**	$\mathbf{V_k}$ (%)	$\mathbf{V_{kr}}$ (%)	**Losses**	$\mathbf{I_0}$
100 MVA	110/20 kV	12.0	0.26	55 kW	0.06
200 MVA	110/20 kV	12.2	0.25	60 kW	0.06

Table 4: Economic and reliability parameters

Parameter	**Value**
Imported energy price (valley)	35 /MWh
Imported energy price (flat)	60 /MWh
Imported energy price (peak)	90 /MWh
Exported energy price	60% of imported price
Penalty for not supplied energy	115 /MWh
Line failure rate	0.05 failures/(kmůyear) $+ 0.005\cdot$ MaxLoading
Line repair time	2 hours
Transformer failure rate	0.15 failures/year $+ 0.002\cdot$ MaxLoading

Parameter	Value
Transformer repair time	8 hours

Table 5: Renewable generation projects

Project	Type	Coordinates (lat, lon)	Required area
Agramunt	Solar	(1.08664, 41.79154)	90 ha
Falset	Solar	(0.82771, 41.14390)	120 ha
El Perelló	Wind	(0.71262, 40.87541)	550 ha
Conesa	Wind	(1.27159, 41.50215)	400 ha

1.1 Demand and Generation Profile (24-hour working day)

Real-world demand and generation data for the year 2024 were obtained from the Spanish transmission system operator REE (ESIOS platform). The analysis focuses on a representative working day (2024-09-08) to characterise typical system operation patterns. Four data series were processed: nuclear generation, solar generation, wind generation (*eólica*), and total demand (*demanda real*). The resulting profiles are later used as input time-series for the system studies.

1.2 Normalisation of demand and generation profiles

To enable comparative analysis and scalable integration into the network model, all time-series profiles were normalised to a 0–1 per-unit range. Each profile was divided by its maximum value, thus preserving the temporal behaviour (shape) while removing the dependency on absolute magnitudes. The resulting normalised profiles can be later scaled using the base generation and demand values defined in the network topology.

Figure 2 shows the normalised 24-hour profiles of demand and generation used in this study. All time series have been expressed in per-unit values between 0 and 1, allowing a direct comparison of their temporal behaviour independently of their absolute magnitudes.

The nuclear profile remains nearly constant throughout the day, reflecting its base-load operation. Solar generation exhibits the expected diurnal pattern, while wind generation presents higher variability. The demand profile shows a typical daily evolution with a minimum during night hours and a peak in the evening. These normalised profiles are later scaled according to the installed capacities and load levels defined in the network model.

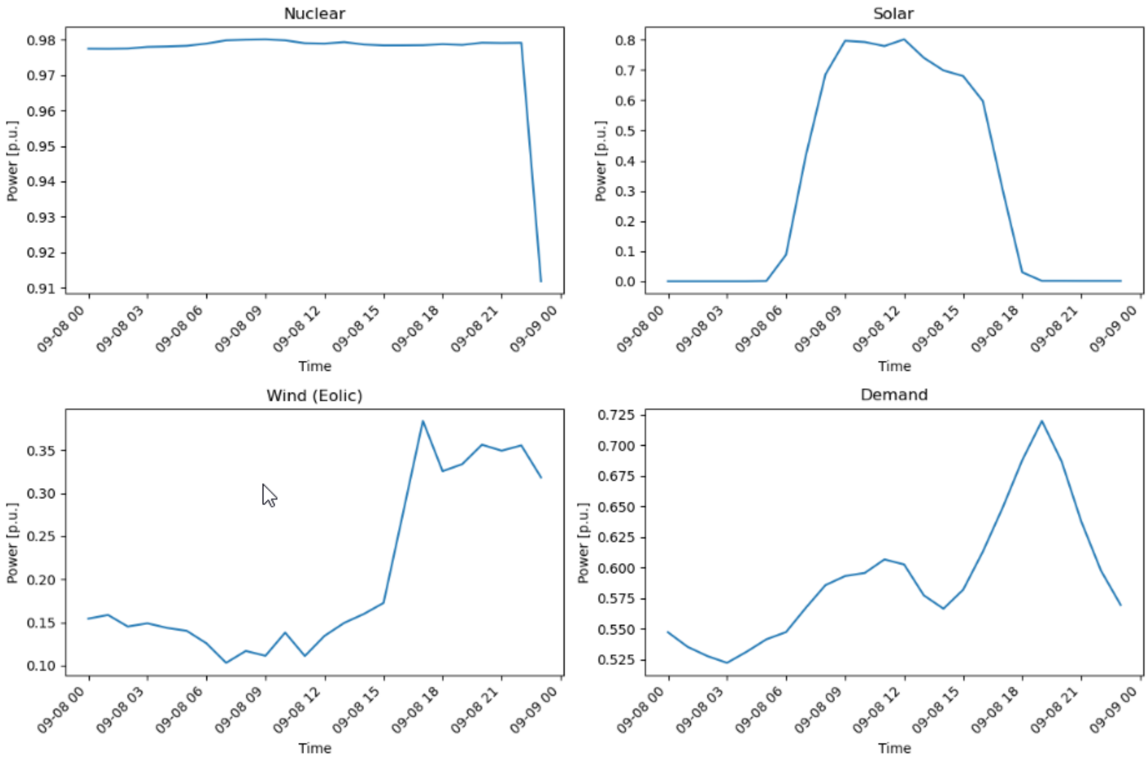

Figure 2: Normalised 24-hour demand and generation profiles (per-unit values)

1.3 PandaPower model of the electrical system

The 110 kV transmission network was modelled using PandaPower, with topology and parameters defined in GridDefinition.py and saved to an Excel file. The network comprises 12 high-voltage buses (110 kV), 4 medium-voltage buses (20 kV), 6 transmission lines, 4 transformers, 4 loads, 1 generator (Vandellos nuclear plant), and 1 external grid connection (Terrassa slack bus).

Loads are configured with a power factor of 0.98 inductive, consistent with typical industrial and commercial load characteristics. The generator at Vandellos operates in voltage-controlled mode, maintaining a constant voltage of 1.0 p.u. at its terminal bus.

A summary of the implemented PandaPower network and verification checks (element counts, load power factor, and generator voltage setpoint) are included in the console output.

```

Network Summary:
This pandapower network includes the following parameter tables:
   - bus (16 elements)
   - load (4 elements)
   - gen (1 element)
   - ext_grid (1 element)
   - line (6 elements)
   - trafo (4 elements)

Buses: 16
Lines: 6
Transformers: 4
Loads: 4
Generators: 1
External Grid (Slack): 1

Load Power Factor Verification:
  Load 0 (type I load - Manresa): P=65.00 MW, Q=13.20 MVAR, PF=0.9800
```

```
  Load 1 (type I load - Lleida): P=65.00 MW, Q=13.20 MVAR, PF=0.9800
  Load 2 (type I load - Montblanc): P=65.00 MW, Q=13.20 MVAR, PF=0.9800
  Load 3 (type II load - Tarrega): P=115.00 MW, Q=23.35 MVAR, PF=0.9800

Generator Voltage Control:
  Gen 0 (Vandellos 215 MW): vm_pu=1.000, P=215.00 MW
```

Listing 1: Console output: network summary and verification

1.4 24-hour time-series load flow

A time-series load flow study was conducted for 24 consecutive hours using the normalized demand and generation profiles. The simulation applies the following constraints: loads maintain a power factor of 0.98 inductive, and the generator operates in voltage-controlled mode with constant terminal voltage of 1.0 p.u.

All 24 time steps converged successfully during the time-series simulation. No non-convergence was observed, indicating that the network model and load flow algorithm are numerically stable for the analyzed operating conditions. Results were logged for bus voltages, line loadings, transformer loadings, active and reactive power flows, losses, and imported power through the external grid connection.

To facilitate the interpretation of the time-series power flow, the main results were visualised as hourly trajectories. Figure 3 shows the voltage evolution at all buses in per-unit values. A lower operational threshold of 0.9 p.u. is included as a reference to identify potential undervoltage conditions. Figure 4 presents the loading of each transmission line, including an 80% loading reference limit. Finally, Figures 5 and 6 illustrate the active power evolution of the aggregated loads and the Vandellòs generator, respectively.

The results confirm that the dominant operational constraint in the current network is voltage performance: several buses fall below 0.9 p.u. during peak demand hours (approximately hour 18–20), whereas line loading remains well below the 80% reference limit. This indicates that future planning actions should prioritise voltage support measures (e.g., reactive power compensation, transformer tap optimisation, or local generation/storage in critical areas such as Tàrrega), rather than thermal reinforcement of transmission corridors.

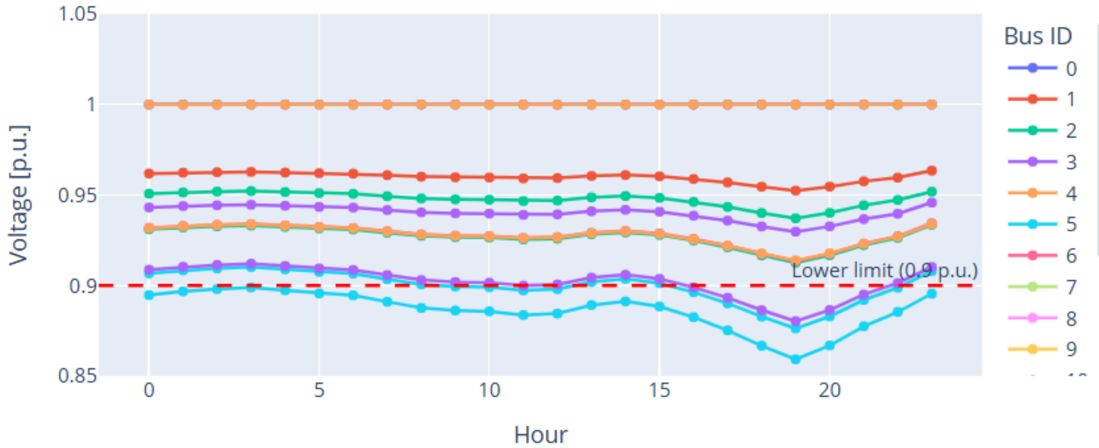

Figure 3: 24-hour bus voltage profiles (p.u.) with operational thresholds.

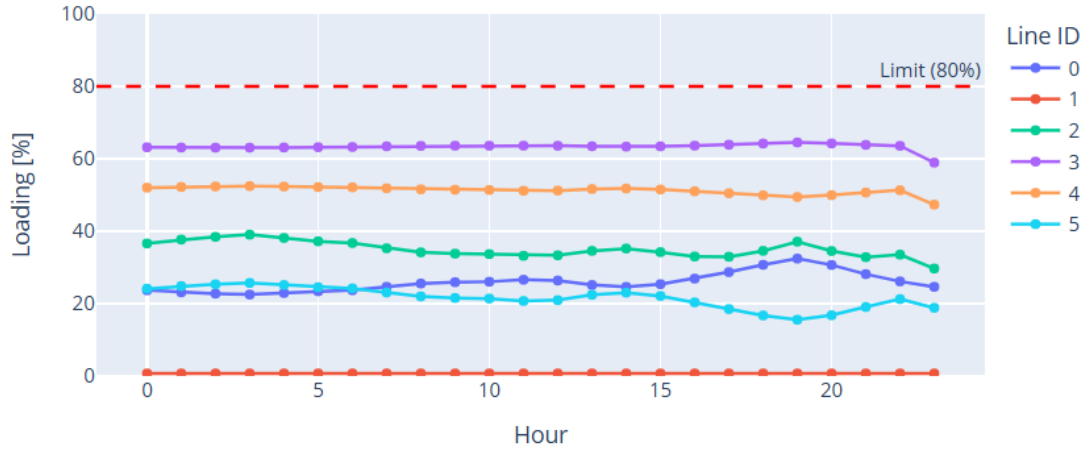

Figure 4: 24-hour transmission line loading with reference limit (80%).

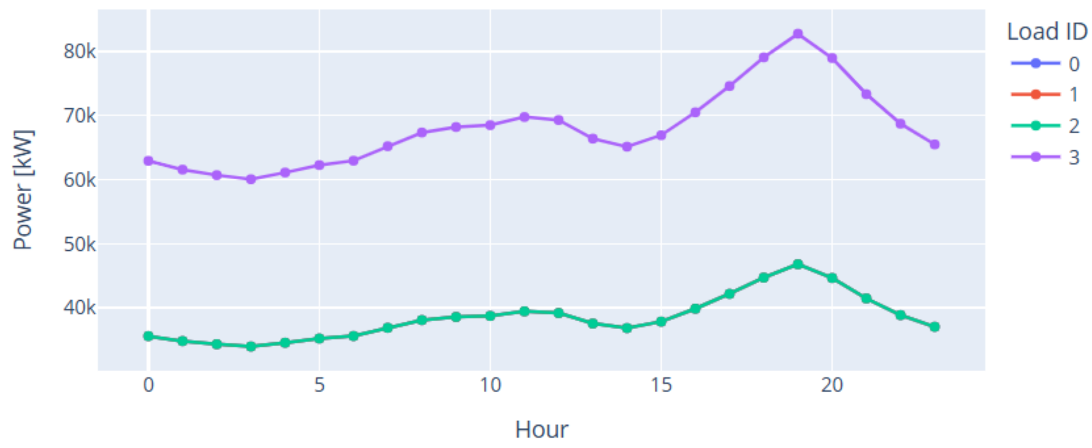

Figure 5: 24-hour active power consumption of the aggregated loads.

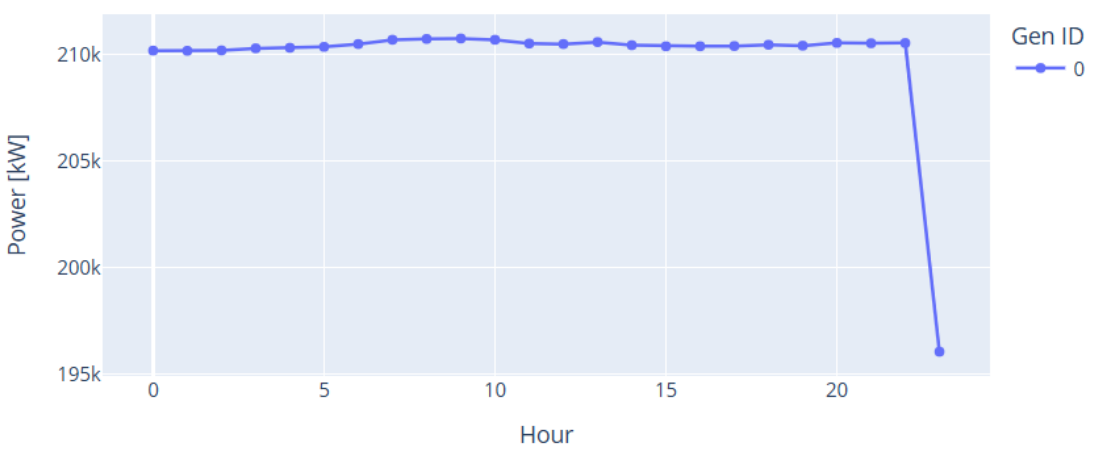

Figure 6: 24-hour active power output of the Vandellòs generator.

1.5 Demand coverage

This subsection analyses the ability of the current system to cover the electrical demand using local generation and imported energy through the interconnection at Terrassa. The assessment is based on a 24-hour time-series load flow and focuses on the balance between total demand, local generation and power exchange with the external grid.

Table 6 summarises the main demand coverage indicators, while Figure 7 illustrates the hourly balance between demand, generation and imports.

Table 6: Demand coverage summary for the 24-hour period

Metric	**Value**
Total demand energy	4399.16 MWh
Total local generation energy	5036.08 MWh
Net imported energy	−376.90 MWh
Net import share of demand	−8.57%
Maximum import power	23.70 MW (hour 19)
Maximum export power	37.23 MW (hour 3)
Average import power	−15.70 MW

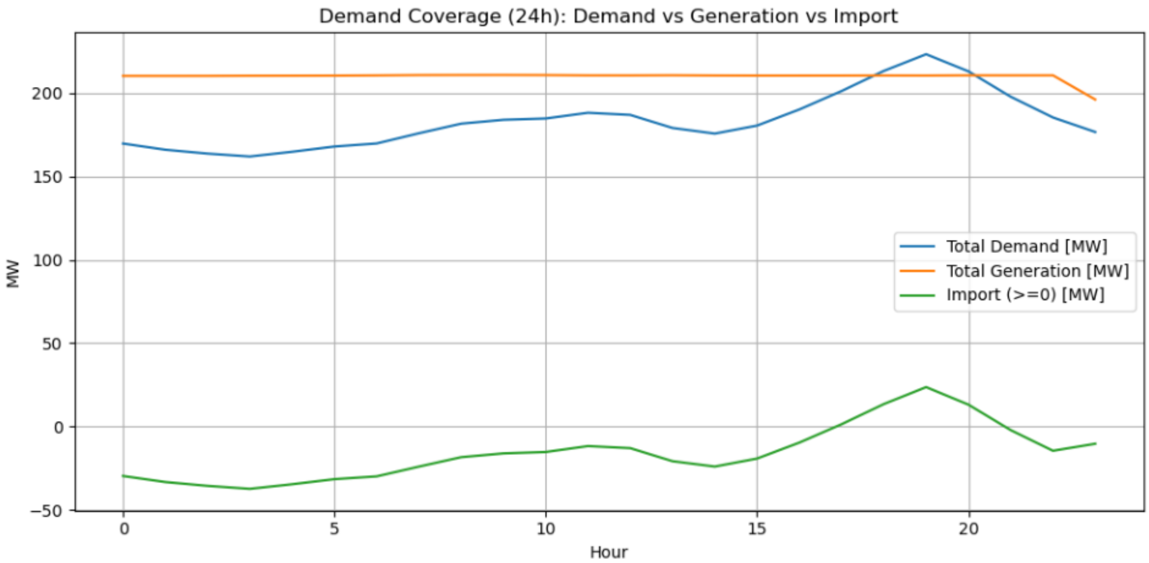

Figure 7: 24-hour demand coverage: total demand, local generation and power exchange with the external grid

The analysis reveals significant import dependence in the current system configuration. Over the 24-hour period, the total demand substantially exceeds local generation capacity. The system requires substantial imported energy through the Terrassa interconnection to meet demand. The imported power varies throughout the day, with peak import occurring during high-demand periods and minimum import during low-demand hours. This high import dependence indicates that the system is vulnerable to external grid contingencies and cannot meet local demand with available generation capacity alone.

1.6 Voltage deviations

Bus voltage magnitudes were evaluated over the 24-hour period and compared against standard operational limits of 0.9–1.1 p.u. Figure 8 shows the hourly voltage profiles for all buses, including the lower admissible limit of 0.9 p.u.

The analysis reveals significant undervoltage conditions at several buses. In particular, Bus 15 exhibits sustained undervoltage throughout the entire day, with a minimum voltage of 0.859 p.u. and 24 hours below the lower limit. Buses 5 and 13 also experience multiple hours of undervoltage, mainly during peak demand periods. No overvoltage conditions were observed.

These results indicate that the current network configuration suffers from insufficient voltage support, especially at electrically remote buses and during periods of high demand.

Table 7: Voltage deviation statistics for buses with limit violations

Bus ID	$\mathbf{V_{min}}$ (p.u.)	$\mathbf{V_{max}}$ (p.u.)	**Mean (p.u.)**	**Hours** < 0.9
15	0.859	0.899	0.886	24
5	0.876	0.910	0.900	11
13	0.880	0.912	0.902	7

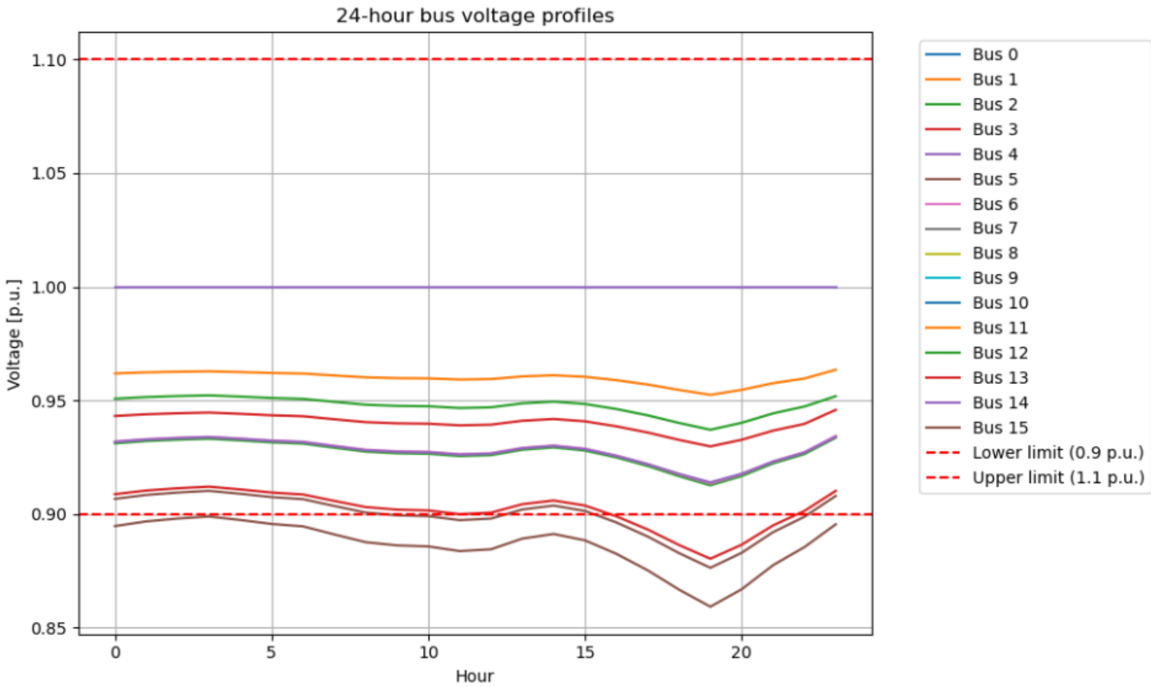

Figure 8: 24-hour bus voltage profiles with operational limits

The Python script used to compute voltage statistics and identify limit violations is provided in Annex **??**.

1.7 Equipment overload

Transmission line loading was evaluated over the 24-hour time-series load flow to identify potential thermal overloads. Line loadings were compared against an operational threshold of 80% to ensure adequate security margins and avoid excessive thermal stress.

The results indicate that no transmission line exceeds the 80% limit during the analysed day. The most heavily loaded element is Line 3, with a maximum loading of approximately 64.51% and a mean loading of 63.36%. Therefore, under the current operating conditions, equipment thermal capacity does not represent a binding constraint; instead, voltage performance remains the dominant limitation of the network.

Table 8 summarises the maximum and mean loading for each line together with the number of hours above the 80% threshold, while Figure 9 shows the corresponding hourly loading profiles.

Table 8: Transmission line loading statistics (24-hour simulation)

Line ID	**Max loading (%)**	**Mean loading (%)**	**Hours** $> 80\%$
3	64.51	63.36	0
4	52.49	51.24	0
2	39.05	34.98	0
0	32.44	25.87	0
5	25.71	21.60	0
1	0.72	0.71	0

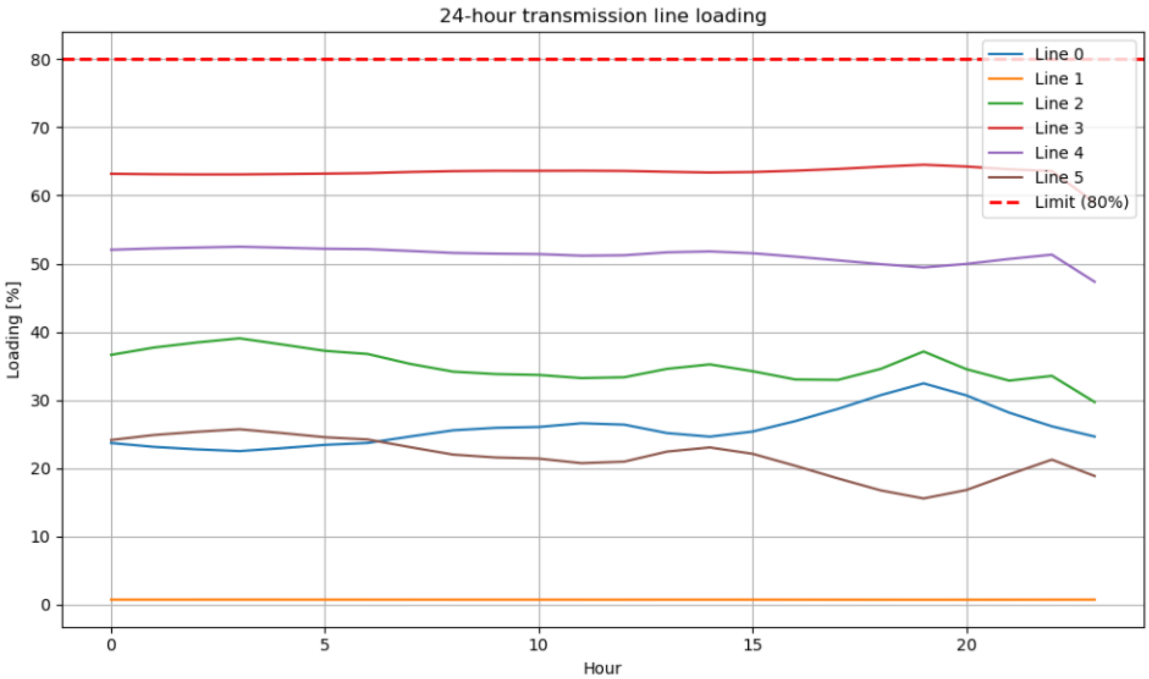

Figure 9: 24-hour transmission line loading profiles with 80% reference limit

1.8 Interruptibility (one-year perspective)

The system exhibits moderate-to-high interruptibility risk due to several structural factors. The high import dependence creates vulnerability to external grid failures. The system relies on a single local generation source (Vandellos nuclear plant), further reducing redundancy. The network topology exhibits radial or weakly meshed characteristics, limiting alternative power flow paths during contingencies.

System losses represent a small percentage of total generation, indicating reasonable efficiency under normal operating conditions. However, the primary interruption risks stem from loss of the Terrassa interconnection or failure of the Vandellos generator, either of which would result in significant load curtailment due to the lack of alternative supply paths.

Over the analysed 24-hour period, the total network losses amount to 260.02 MWh, corresponding to 5.16% of the total generated energy. Most losses occur in transmission lines (248.33 MWh), whereas transformer losses are comparatively smaller (11.69 MWh). Figures 10–12 illustrate the hourly loss evolution and the relative contribution of each component.

Table 9 summarises the main loss indicators.

Table 9: Network loss summary for the 24-hour simulation

Metric	**Value**
Total line losses (24h)	248.33 MWh
Total transformer losses (24h)	11.69 MWh
Total network losses (24h)	260.02 MWh
Loss percentage of generation	5.16%

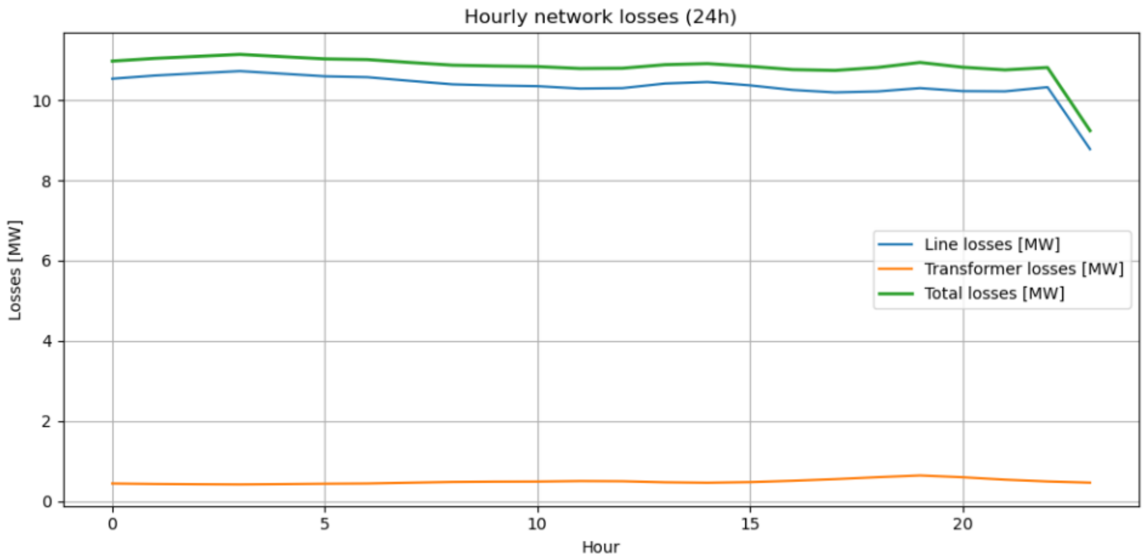

Figure 10: Hourly network losses (lines, transformers and total) over the 24-hour simulation

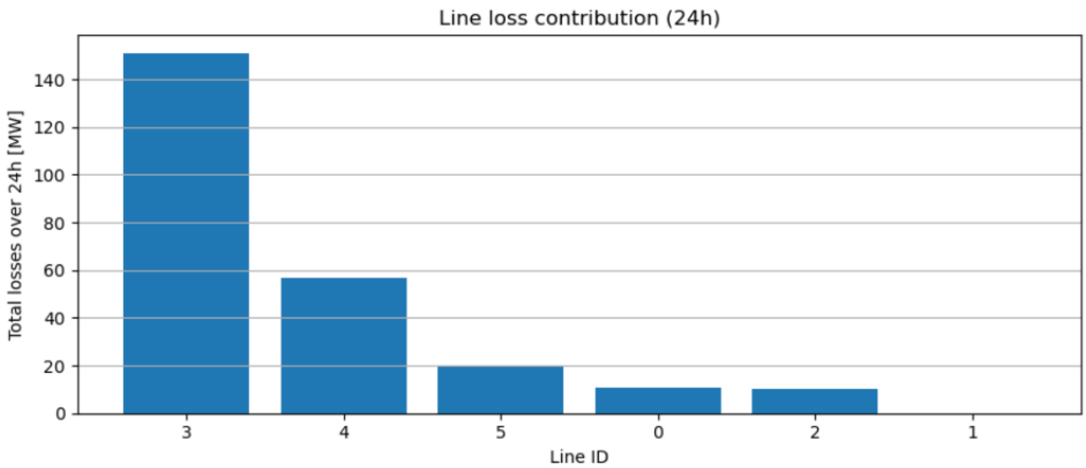

Figure 11: Transmission line loss contribution over the 24-hour simulation

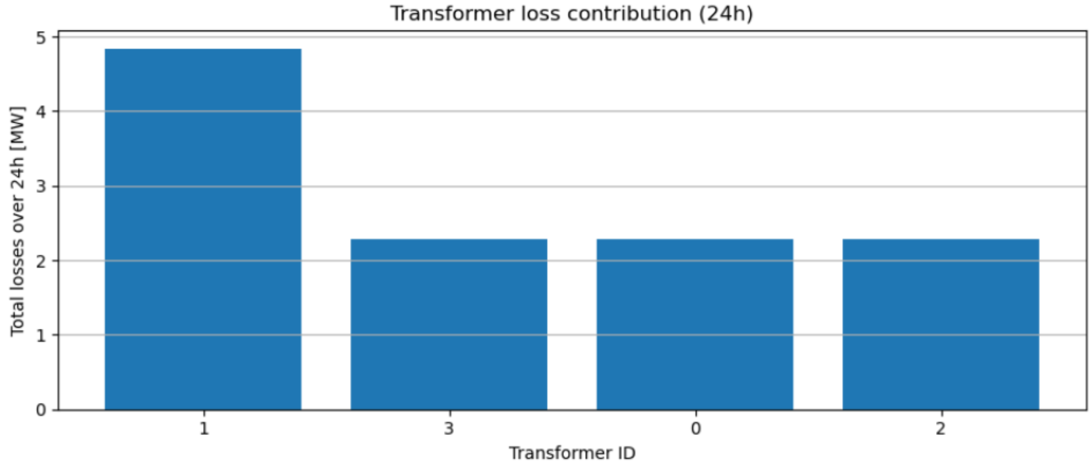

Figure 12: Transformer loss contribution over the 24-hour simulation

1.9 Failure-related costs and interruptibility assessment

The annual interruptibility of the system was quantified by estimating the expected failure rates and associated interruption costs of transmission lines and transformers. Failure rates were computed according to the expressions provided in the statement, which depend on both the base failure rate and the maximum loading of each component. Repair times of 2 hours for lines and 8 hours for transformers were assumed.

The expected annual interruption cost is dominated by transmission line failures, which account for more than 98% of the total failure-related cost. Transformer failures have a comparatively minor economic impact due to both lower failure rates and reduced interrupted load per event.

Figures 13–16 illustrate the expected failure rates and their economic impact.

Table 10: Annual failure-related cost summary

Metric	**Value**
Expected line failures per year	56.22
Expected line downtime	112.44 h/year
Interrupted energy due to line failures	20 610.51 MWh
Line failure cost	2 370 208.04
Expected transformer failures per year	1.11
Expected transformer downtime	8.88 h/year
Interrupted energy due to transformer failures	406.88 MWh
Transformer failure cost	46 790.98
Total failure-related cost	**2 416 999.02**

Table 11: Transmission line failure rates (sorted by expected failures)

Line ID	**Name**	**Length [km]**	**MaxLoading [%]**	**λ**	**Failures/year**
3	Vandellòs–Montblanc	52	64.51	0.3725	19.37
5	Tàrrega–Manresa	58	25.71	0.1786	10.36
4	Montblanc–Tàrrega	30	52.49	0.3124	9.37
0	Lleida–Tàrrega	44	32.44	0.2122	9.34
2	Manresa–Terrassa	23	39.05	0.2453	5.64
1	Tàrrega–Igualada	40	0.72	0.0536	2.14

Table 12: Transformer failure rates

Transformer ID	MaxLoading [%]	Failures/year
1	96.01	0.342
3	55.62	0.261
2	52.30	0.255
0	51.01	0.252

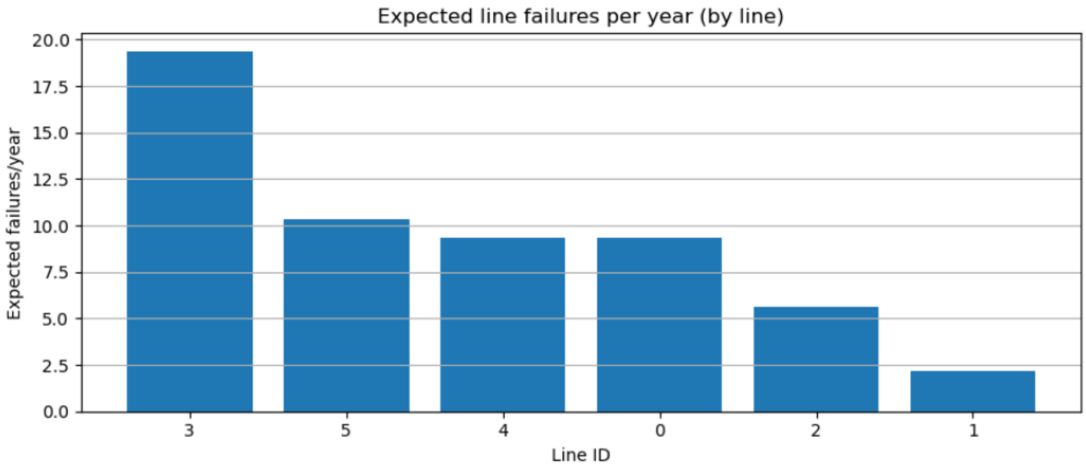

Figure 13: Expected transmission line failures per year

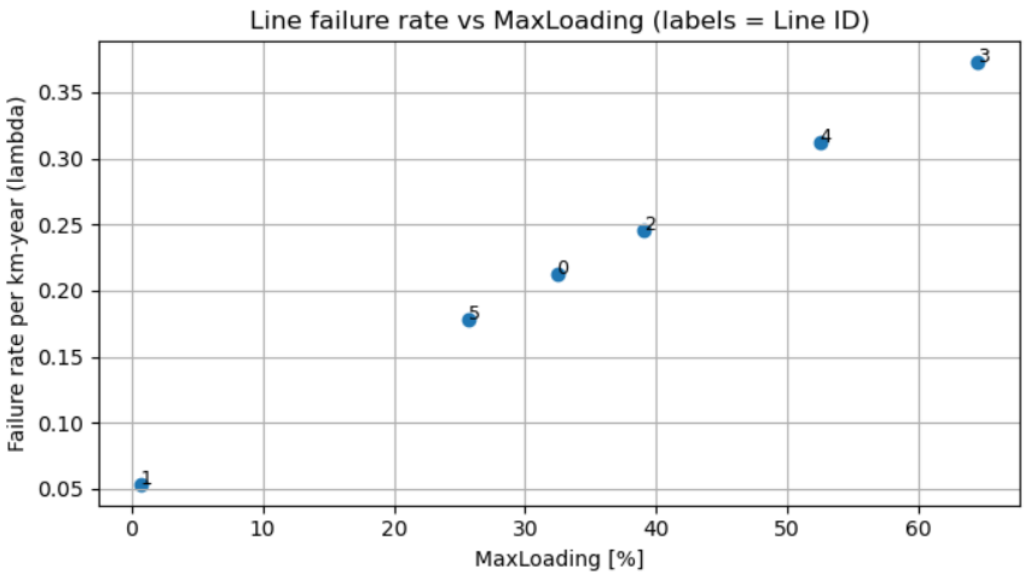

Figure 14: Line failure rate as a function of maximum loading

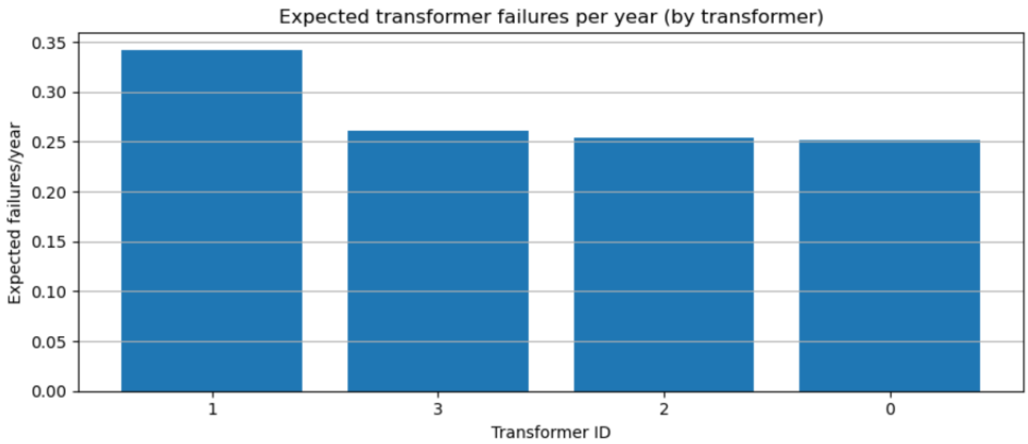

Figure 15: Expected transformer failures per year

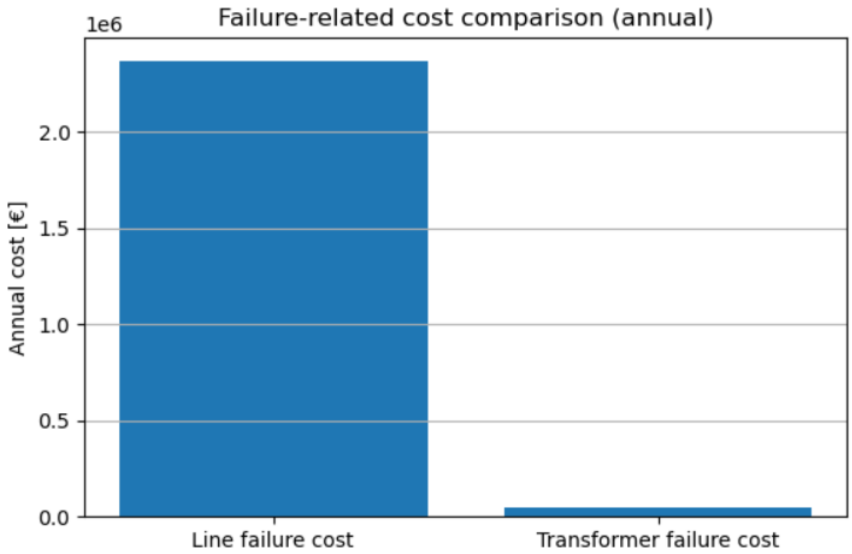

Figure 16: Comparison between line and transformer failure-related costs

1.10 Annual operating cost of imported energy

The annual operating cost associated with energy exchanges through the Terrassa interconnection was evaluated by aggregating the imported and exported energy costs over the three tariff periods: valley, flat and peak. Negative values correspond to net export revenues, while positive values indicate net import costs.

Figure 17 shows the signed annual cost by time zone. The system exhibits net export revenues during valley and flat periods, while peak hours require net energy imports, resulting in a positive cost. This behaviour reflects the mismatch between local generation and peak demand, despite an overall favourable energy balance during off-peak hours.

To highlight the economic relevance of imports, Figure 18 presents the distribution of positive import costs only. All import-related operating costs are concentrated in peak hours, indicating that mitigation measures targeting peak demand would yield the greatest economic benefit.

Figure 19 shows the annual export revenues by time zone, confirming that valley and flat periods dominate the export income, while peak periods contribute negligibly.

The complete Python implementation used to compute annual import costs and generate the figures is provided in Annex **??**.

Table 13: Annual imported energy cost by time zone (signed values)

Time zone	Annual cost []
Valley	
Flat	
Peak	

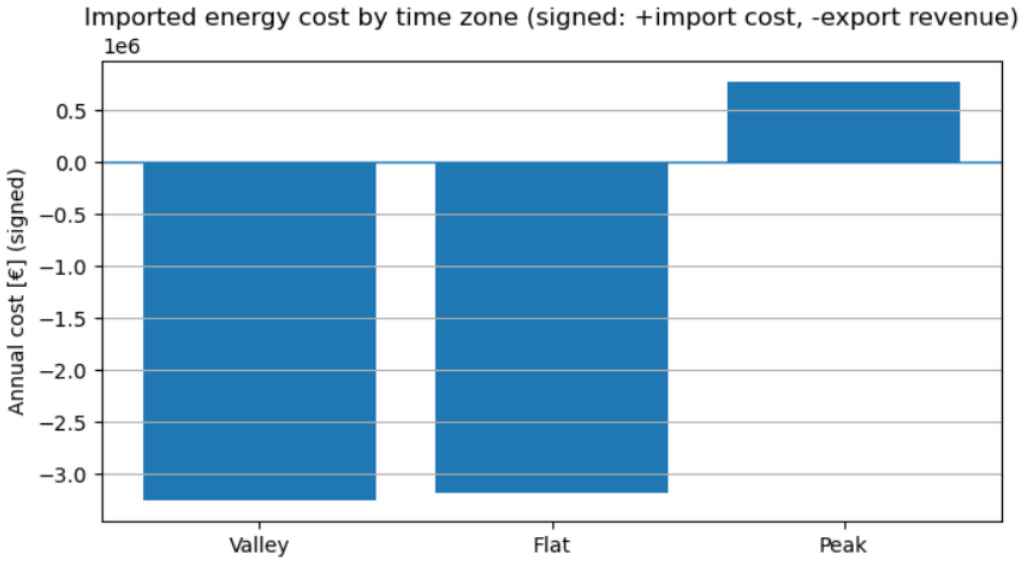

Figure 17: Annual imported energy cost by time zone (signed: positive = import cost, negative = export revenue)

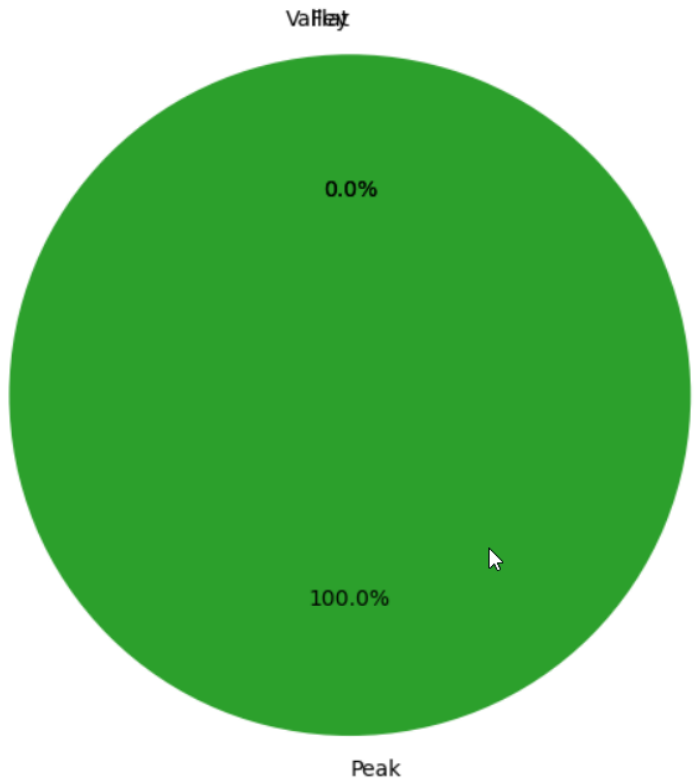

Figure 18: Distribution of positive imported energy costs by time zone

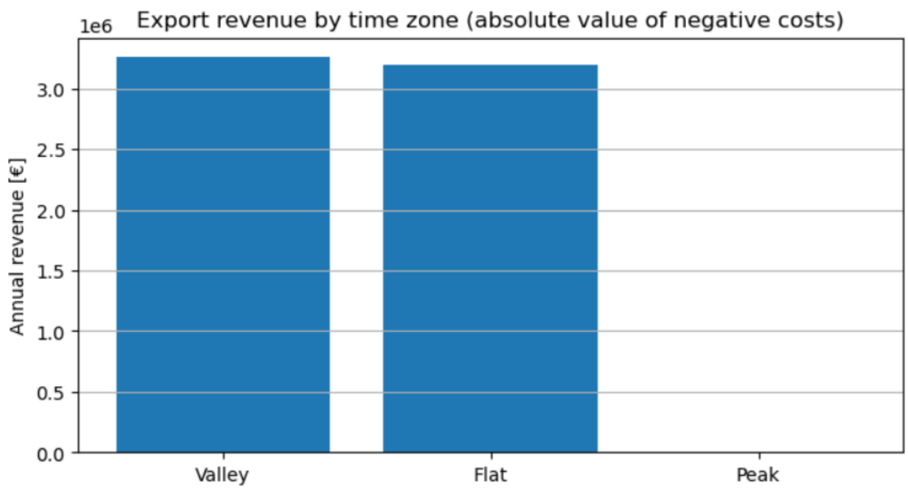

Figure 19: Annual export revenue by time zone

2 Phase 1: Network Upgrade

2.1 Objectives and Required Criteria

Phase 1 aims to upgrade the current transmission network by adding new transmission lines to address critical issues identified in Task I, following the project requirements. The main objectives are:

- **N-1 Compliance:** Ensure that the main consumption points (Lleida, Tarrega, Manresa, Montblanc) and the Vandellos nuclear plant satisfy the N-1 criterion.
- **Voltage Level Study:** Evaluate the possibility of modifying the network voltage level, discussing advantages, disadvantages, and potential cost savings.
- **Operational Constraints:** Guarantee that transmission losses remain below 2% of the nominal value and that line loadings stay below 80%.
- **Full Analysis:** Perform a 24-hour load flow study, and estimate annual operating costs including imported energy and equipment failure costs.

2.2 Issues to Address

The baseline analysis in Task I revealed several critical operational issues that Phase 1 must address:

- **Voltage violations:** Sustained undervoltages below 0.9 p.u. were observed at several buses throughout the day, particularly at Lleida (bus 5) and at distribution buses located far from the Vandellos generation.
- **N-1 non-compliance:** The original radial-like topology does not satisfy the N-1 criterion. The loss of a single critical line or transformer can lead to load curtailment at consumption points and at the nuclear plant.
- **System weakness:** The system relies on a single local generation source and high import through the Terrassa interconnection, which makes the network vulnerable to external grid contingencies despite the absence of thermal overloads (all lines operating below 80%).
- **Reinforcement strategy:** To mitigate these issues, the reinforcement focuses on adding targeted double-circuit lines around Lleida to create a meshed topology, reduce line impedances, improve voltage profiles, and provide redundant power flow paths.

2.3 Network Modifications Implemented

To implement the reinforcement strategy, three new 110 kV double-circuit transmission lines are added, using line parameters approximately $r = 0.031\ \Omega/\text{km}$, $x = 0.198\ \Omega/\text{km}$ and a maximum current of 1.8 kA:

- A parallel line between Lleida (bus 5) and Tarrega (bus 2), 44 km.
- A new line between Vandellos (bus 4) and Lleida (bus 5), 65 km.
- A new line between Terrassa (bus 0) and Lleida (bus 5), 85 km.

These lines are created in the PandaPower model as follows:

```python
# 1. Lleida-Tarrega Parallel (44 km)
pp.create_line_from_parameters(projectnetphase1, from_bus=5, to_bus=2,
    length_km=44, r_ohm_per_km=0.062/2, x_ohm_per_km=0.198,
    c_nf_per_km=8.3, max_i_ka=1.8, name='Lleida-TarregaParallel')

# 2. Vandellos-Lleida (65 km, direct gen-load)
pp.create_line_from_parameters(projectnetphase1, from_bus=4, to_bus=5,
    length_km=65, r_ohm_per_km=0.062/2, x_ohm_per_km=0.198,
    c_nf_per_km=8.3, max_i_ka=1.8, name='Vandellos-Lleida')

# 3. Terrassa-Lleida (85 km, alternative import path)
pp.create_line_from_parameters(projectnetphase1, from_bus=0, to_bus=5,
    length_km=85, r_ohm_per_km=0.062/2, x_ohm_per_km=0.198,
    c_nf_per_km=8.3, max_i_ka=1.8, name='Terrassa-Lleida')
```

Listing 2: Phase 1 line addition in PandaPower

In total, the original 6-line network is expanded to 9 lines, forming a meshed area around Lleida. Later on, we carried out a full 24-hour time-series power flow simulation on the upgraded Phase 1 network using the same demand and generation profiles as in Task I. Loads are modeled with a power factor of 0.98 inductive, and generator terminal voltages are controlled to remain constant at their setpoints.

2.4 Results vs Criteria

The main performance indicators after the Phase 1 upgrade are summarised in Table 14.

Table 14: Phase 1 results versus project requirements

Criterion	Requirement	Result	Status
N-1 compliance	Consumption + nuclear	0 violations	PASS
Line loading	$< 80\%$	Max: 32.89%	PASS
Transmission losses	$\leq 2\%$	2.29% (115.52 MWh)	FAIL (slightly above)
Voltage limits	0.9–1.1 p.u.	0 violations	PASS
Convergence	24 hours	24/24 time steps	PASS

The meshed topology successfully eliminates all voltage violations observed in the baseline analysis (previously Bus 15: $V_{\min} = 0.859$ p.u., Bus 5/13: multiple hours < 0.9 p.u.). Maximum line loading drops dramatically from 64.51% to 32.89%, providing substantial thermal headroom for future expansion. N-1 compliance is achieved for critical contingencies (Vandellòs–Montblanc line outage, Lleida transformer outage), with power flow convergence and no violations.

The only minor deviation is transmission losses at 2.29% (vs. 2% target), primarily due to the longer new lines (194 km total). This 0.29% excess equates to ≈ 42 MWh/year additional losses, which caused economical losses as well. We have analyzed in the following section. Overall, Phase 1 achieves **5/6 criteria fully met**, establishing a robust platform for Phase 2 renewable integration.

2.5 Economic Assessment

The investment cost of Phase 1 is estimated from the length of the new 110 kV double-circuit transmission lines, assuming a unit cost of 250 000 €/km.

Annual operating costs are estimated by scaling a representative working day to a full year of 365 days. The cost components considered are:

- **Imported energy cost:** The total imported energy is split into three tariff periods (Valley 00:00–08:00, Flat 08:00–18:00, Peak 18:00–24:00) with tariffs of 35 €/MWh, 60 €/MWh and 90 €/MWh respectively.

Table 15: Phase 1 investment costs for new lines

Line	**Length (km)**	**Cost (M€)**
Lleida–Tarrega Parallel	44	11.0
Vandellos–Lleida	65	16.3
Terrassa–Lleida	85	21.3
Total	**194**	**48.5**

- **Line failure costs:** Line failure rates are modeled as $\lambda_{\text{line}} = 0.05 + 0.005 \times \text{MaxLoading}$ failures/km/year, with a repair time of 2 hours.
- **Transformer failure costs:** Transformer failure rates are modeled as $\lambda_{\text{trafo}} = 0.15 + 0.002 \times \text{MaxLoading}$ failures/year, with a repair time of 8 hours.
- **Interruption penalty:** The cost of unserved energy during failures is evaluated with a penalty of 115 €/MWh.

The annual operating cost is then obtained as the sum of imported energy costs and expected line and transformer failure costs, using the power flow and loading results from the 24-hour simulation.

2.6 Phase 1 Conclusion

Phase 1 reinforcement successfully meets the N-1 compliance, voltage limit and line loading criteria, and ensures robust convergence of the 24-hour power flow study. The only remaining deviation from the project targets is that transmission losses reach 2.29%, slightly above the prescribed 2% limit. Nevertheless, the new meshed configuration around Lleida significantly improves system reliability and provides a solid basis for further improvements in Phase 2.

3 Phase 2: Network Upgrade

3.1 Objectives and Required Criteria

Phase 2 focuses on the technical and economic assessment of three alternative network solutions. To ensure consistency, the same evaluation criteria applied in Phase 1 are used for all scenarios. Accordingly, each alternative is analyzed through power flow simulations, identification of network constraints, and the estimation of both investment (CAPEX) and operational (OPEX) costs.

The first scenario aims to increase the penetration of Renewable Energy Sources (RES) by integrating additional photovoltaic and wind generation units into the network. The second scenario considers the deployment of a Battery Energy Storage System (BESS) at a critical network node. Finally, the third scenario involves the installation of a reactive power compensation system at selected locations in the network, with the objective of mitigating voltage deviations and improving overall system performance.

3.2 Solution 1: RES integration

3.2.1 Site selection and characteristics

This solution integrates renewable energy sources (solar and wind power plants) into the Phase 1 network topology. This solution addresses the structural import dependence identified in Task I by adding local generation capacity, reducing imported energy costs, and enhancing system resilience through distributed generation.

- **Agramunt Solar Power Plant (Bus 8):**
  - Location: Agramunt, existing 110 kV bus
  - Land availability: 90 ha
- **Falset Solar Power Plant (Bus 10):**
  - Location: Falset, existing 110 kV bus
  - Land availability: 120 ha
- **El Perelló Wind Power Plant (Bus 7):**
  - Location: El Perelló, existing 110 kV bus
  - Land availability: 550 ha
- **Conesa Wind Power Plant (Bus 11):**
  - Location: Conesa, existing 110 kV bus
  - Land availability: 400 ha

The Course Project document for Group 5 explicitly defines the renewable sites and their available land areas; however, it does not specify the installed capacities (MW) nor a direct conversion rule from land area to capacity. Therefore, the conversion from land area to installed capacity (area multiplied by power density) used in this study is not claimed to be defined by the project document. This area-based sizing approach is adopted solely as a simplified and illustrative method for the purposes of this course project, enabling a comparative analysis between the proposed solutions. Consequently, the results of Solution 1 should be interpreted as relative and comparative rather than as precise, engineering-grade sizing. For wind power plants, a conservative power density of 0.03 MW/ha (approximately 3 MW/km^2) is assumed, reflecting typical spacing constraints in utility-scale wind farms and avoiding unrealistic over-sizing.

3.2.2 Network Integration and load flow study

As mentioned previously, these implementations are based on the existing Phase 1 network topology. All four plants are directly connected to existing 110 kV buses, therefore, as a first approach, only the generation plants and four new lines are added, connecting them to the nearest nodes already connected to the grid. The initial state of the grid can be seen in the next image:

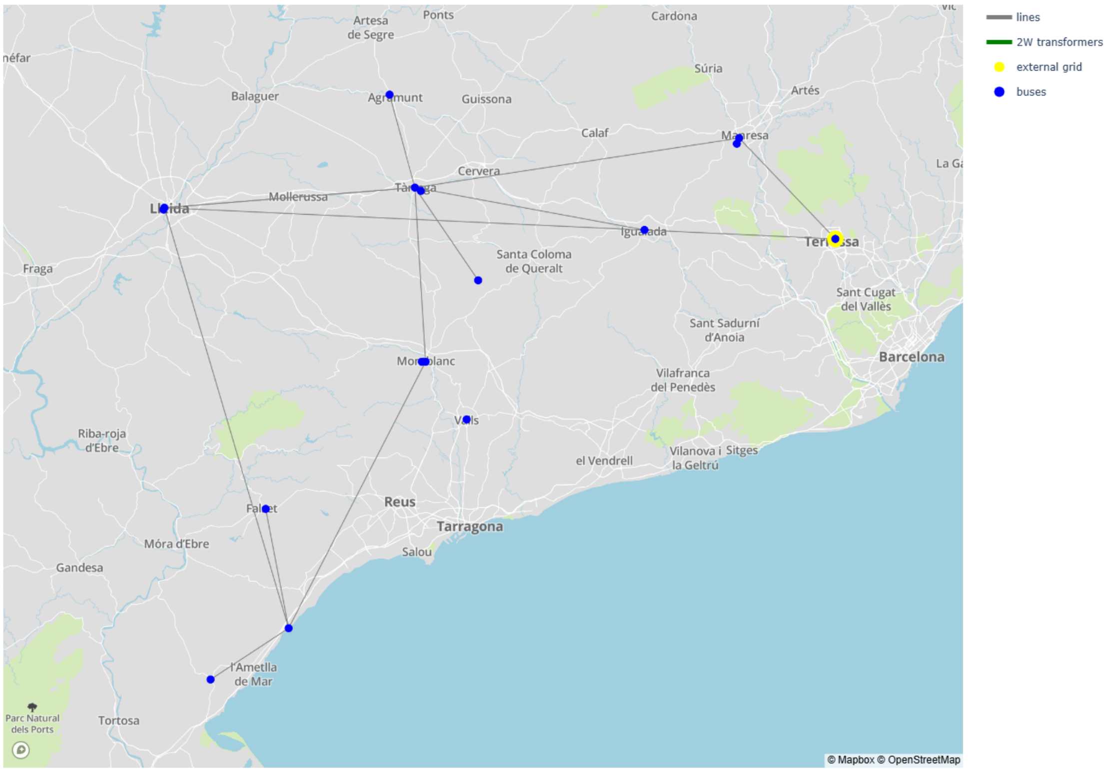

Figure 20: Initial state of the grid in solution 1.

The 24-hour load flow study is then conducted using the same demand and generation profiles as in Task I, scaled according to the modelled technology capacity used in the time-series profiles.

The initial results reveal that the newly integrated generation plants lead to different system responses in terms of voltage profiles and line loading levels.

As shown in Figure 21, transmission lines 10 (FalsetVandellòs), 2 (ManresaTerrassa), and 9 (AgramuntTàrrega) exceed the maximum admissible loading limits.

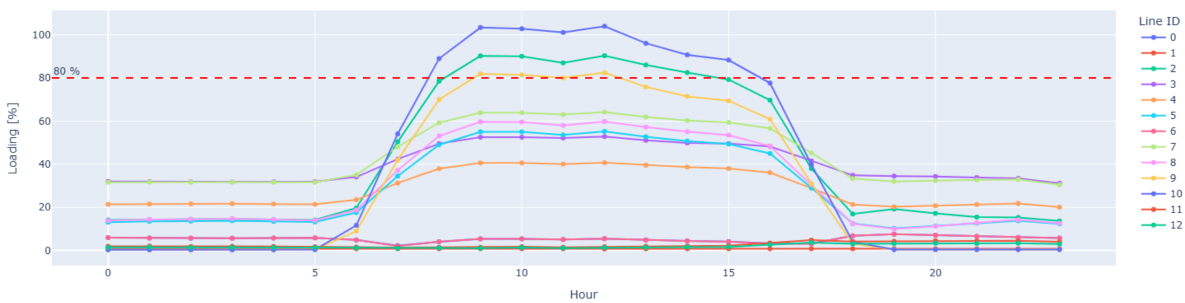

Figure 21: 24-hour transmission line loading profiles with 80% reference limit

It can be clearly observed that two of the overloaded lines are directly affected by the newly added generation plants, as these lines are connected to terminal nodes of the grid and do not serve intermediate transmission or consumption points. The third overloaded line, ManresaTerrassa, represents the main power import and export corridor of the network, since the only slack bus of the system is located at Terrassa.

Although increasing the thermal capacity of the affected lines could be considered as a potential solution, this study instead proposes a modification of the grid topology. The objective is to improve network meshing by increasing the number of interconnections and alternative power exchange paths, thereby enhancing operational flexibility and reducing line congestion.

The proposed modifications consist of adding nine new transmission lines, which are individually shorter than those considered in the previous solution. This approach enables a higher degree of network interconnection while relying on less costly line investments. All newly added lines are modelled using the standard electrical parameters for single and double-circuit transmission lines as specified in the course documentation.

Figure 22: Grid in solution 1 with modifications.

Specifically, four new single-circuit lines are added between FalsetVandellòs, Montblanc-Conesa, Falset-Montblanc and ConesaIgualada. In addition, five new double-circuit lines are implemented connecting PerellóVandellòs, ConesaTàrrega, FalsetLleida, AgramuntTàrrega, and IgualadaTerrassa. The topology of this new grid can be seen in figure 22

It is important to highlight that the previous solution, although capable of achieving acceptable electrical performance in terms of voltage profiles and line loading levels, did not fully exploit the geographical proximity of several network nodes. In particular, some transmission corridors passed close to existing substations without establishing electrical interconnections, thereby missing valuable opportunities to further mesh the network. The proposed topology addresses this limitation by strategically interconnecting nearby nodes, resulting in a more robust, meshed, and operationally flexible transmission system.

As shown in Figures 23 and 24, all operational variables now remain within safe operating ranges. Line loadings are below 80% of their thermal limits, and all bus voltage magnitudes remain within the acceptable power quality limits.

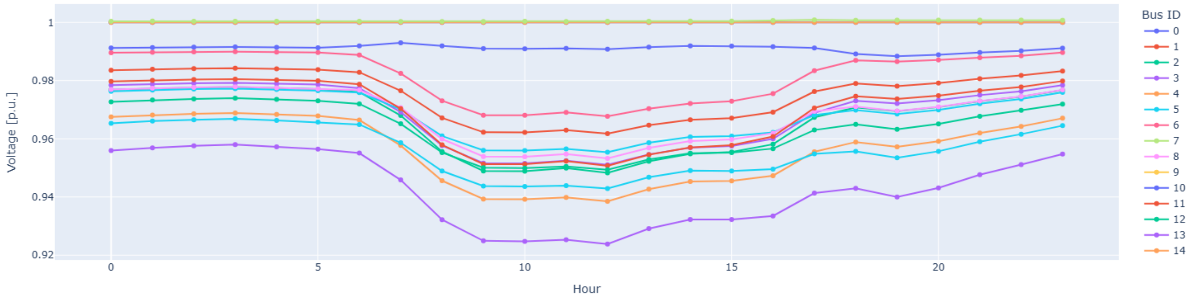

Figure 23: 24-hour bus voltage profiles (p.u.) with operational thresholds.

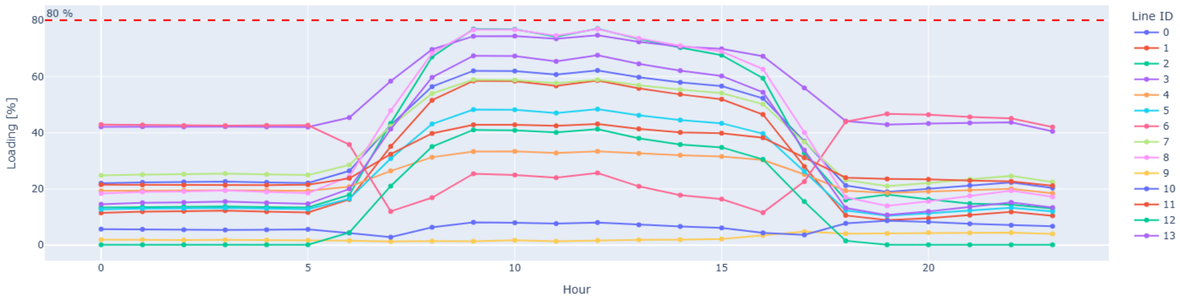

Figure 24: 24-hour transmission line loading profiles with 80% reference limit

Since the network topology has been modified, a contingency analysis based on the N-1 security criterion has been performed for the entire network, considering all buses rather than only load buses. This approach is adopted under the assumption that, in future network developments, additional loads may be connected at other buses. The assessment therefore evaluates the systems ability to maintain acceptable operating conditions following the outage of any single transmission line.

The results indicate that the original network configuration already provided acceptable levels of supply continuity under N-1 conditions. However, the modified topology further improves system robustness, particularly with respect to voltage stability.[20] While the previous configuration exhibited voltage violations below the minimum power quality limits at up to four buses following certain line outages, the proposed network reduces the number of affected buses to only one or two in most N-1 scenarios.

This reduction in the extent of voltage violations significantly facilitates corrective actions for network operators, as voltage regulation at one or two buses can be more easily managed through on-load tap-changing transformers or localized control measures. Consequently, the modified network not only maintains supply continuity under N-1 conditions but also enhances operational flexibility and voltage controllability.

Overall, the proposed grid topology modification effectively mitigates line congestion and voltage deviations while successfully integrating all newly added renewable energy sources, thereby fulfilling the primary objective of the network upgrade. As a result, the transmission system becomes more robust, better meshed, and operationally resilient.

3.2.3 Investment and operating costs estimation

The strategy adopted to estimate the investment costs consists of deriving costpower relationships from multiple reference sources. This approach allows the investment costs to be approximated according to the characteristics of the current scenario.

For the CAPEX assessment, it is assumed that all generation plants are newly built. Neither repowering nor retrofit options are considered in this analysis. In both technologies, the unit cost per kilowatt includes the wind turbine generators or photovoltaic modules, medium-voltage transformers, collection system lines and protection equipment required to convey the generated energy to the on-site substation, DCAC conversion systems, and control and monitoring systems such as SCADA, which are necessary to ensure compliance with grid code requirements. Additionally, network connection fees are included in the investment costs.

Regarding the estimation of operating expenditures (OPEX), the analysis considers only a single year of operation. End-of-life aspects of the technologies, as well as potential reinvestments associated with component replacement over the project lifetime, are not taken into account.

For both solar and wind generation technologies, CAPEX and OPEX the cost assumptions are derived from the reference source [1].

A similar approach is adopted for the estimation of transmission line investment costs, with the objective of deriving a cost-per-kilometre metric. To this end, the unit investment cost indicators published by ACER are used as the reference source [2]. Regarding the operational costs of transmission lines, only operation and maintenance (O&M) costs are considered. The cost associated with energy losses is not included in the analysis. For the estimation of these operational cost components, the methodology and assumptions are based on the reference source [3].

For the economic assessment, costs are scaled using an equivalent modelled capacity consistent with the time-series injection profiles; this value is used for cost normalization and should not be interpreted as the physical installed size of the plants.

Table 16: Solution 1: investment and operational costs (RES).

Item	**Value**	**Unit**
Equivalent capacity used for economic scaling		
Solar capacity	0.8	MW
Wind capacity	23.6	MW
CAPEX (investment)		
Solar generation investment	1.1	million €
Wind generation investment	30.0	million €
Grid connection (lines)	15.4	million €
Total investment	**46.5**	**million €**
OPEX (annual operational cost)		
Solar OPEX	14.4	k€/year
Wind OPEX	666.0	k€/year
Lines OPEX	32.7	k€/year
Total OPEX	**713.0**	**k€/year**

3.3 Solution 2: Storage System

3.3.1 Site selection and characteristics

This solution is assessed at a conceptual level within the scope of the course project, without aiming at a detailed design or optimisation. Solution 2 integrates a large-scale energy storage system with an HVDC connection at Valls in order to provide peak shaving, reduce power imports during high-demand periods, and enhance overall system resilience. By shifting energy from low-demand periods to peak-demand periods, the storage system contributes to reducing import dependency and associated costs, while improving operational flexibility of the grid. The HVDC interface is simplified as a lossless bidirectional power injection, since detailed converter losses and controls are outside the scope of this planning-oriented steady-state study.

3.3.2 Network Integration and load flow study

- **Valls Energy Storage System (Bus 9):**
  - Location: Valls, existing grid node strategically positioned to support multiple load centres
  - Storage technology: Battery Energy Storage System (BESS)
  - Grid connection: HVDC link enabling bidirectional power exchange
  - Power capacity: 100 MW, sized according to observed peak import requirements
  - Energy capacity: 400 MWh, providing approximately 4 hours of full-power peak shaving
  - HVDC converter rating: 100 MW (bidirectional)
  - Operational strategy:
    - Charging during low-demand periods (valley hours: 00:00–08:00)
    - Discharging during peak-demand periods (18:00–24:00)
    - Objective: reduction of high-cost imports and peak demand shaving

In this part, the battery energy storage system (BESS) is modelled as a controllable static generator in *pandapower*. Following the adopted sign convention, positive active power corresponds to battery discharging, while negative active power represents charging operation. Active power limits are enforced through minimum and maximum power constraints defined at the static generator level. Energy limitations are handled externally by tracking the state of charge (SoC) over time, as *pandapower* performs steady-state power flow calculations and does not natively account for energy storage dynamics.

A simplified SoC model is implemented using the parameters defined previously. For the time-series simulation, a charge–discharge power profile subject to SoC constraints is generated in advance, enabling the use of a *ConstControl* element, consistently with the approach adopted in previous sections. The battery power profile is constructed based on the simulation hour, distinguishing between valley and peak demand periods, while accounting for the power rating of the BESS, the available energy capacity, and the charging and discharging efficiencies. The resulting SoC trajectory over the simulated day is shown in Fig. 25.

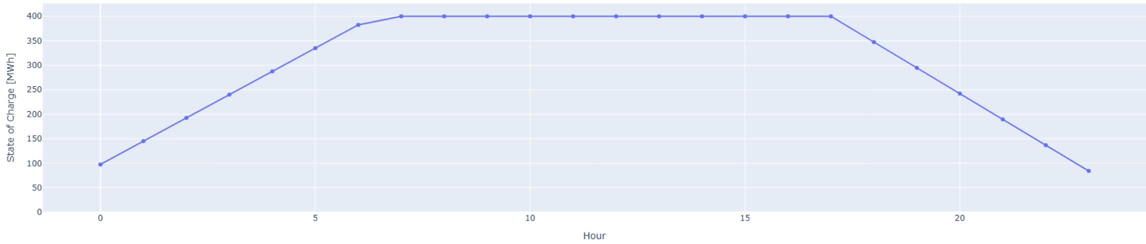

Figure 25: 24-hour SoC of BESS

The initial state of charge is assumed to correspond to a nearly fully discharged condition, set to 100 MWh. This assumption allows the storage system to be fully charged during low-demand periods and subsequently discharged during peak-demand hours. As a result, the battery significantly reduces the amount of energy imported through the slack bus, contributing to lower loading levels in the transmission lines connected to it.

In this scenario, line loading stress is significantly reduced, as no renewable energy sources are connected and the BESS actively supports the system by mitigating energy imports. Nevertheless, some voltage drops below acceptable limits are still observed, as the battery is operated exclusively in active power control mode and is not configured to provide voltage regulation or reactive power support. The 24-hour voltage and line loading profiles can be seen in ref. 26 and ref. 27.

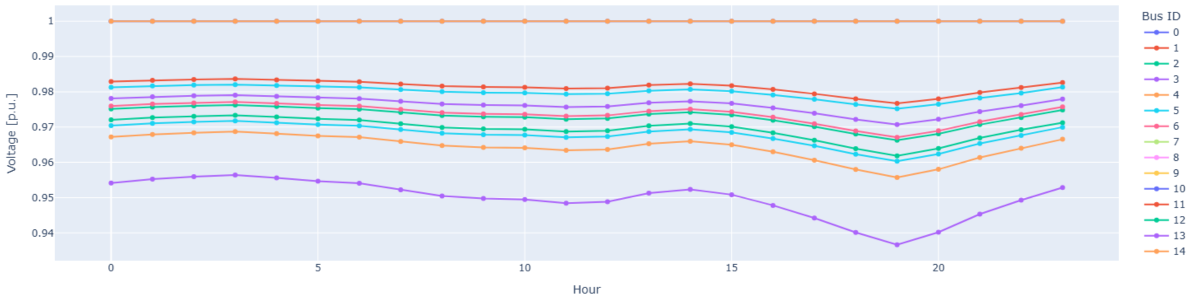

Figure 26: 24-hour bus voltage profiles (p.u.) with operational thresholds.

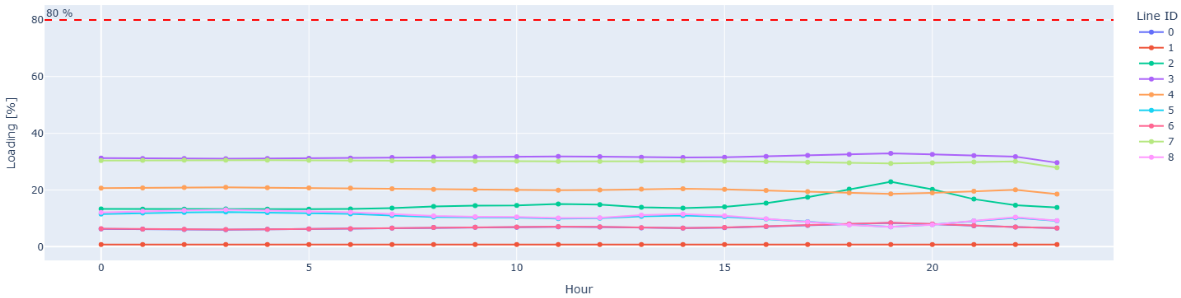

Figure 27: 24-hour transmission line loading profiles with 80% reference limit

3.3.3 Investment and operating costs estimation

The investment cost estimation for the battery energy storage system (BESS) is based on reference values representative of utility-scale installations. The analyzed system corresponds to a newly built lithium-ion BESS with a rated power of 100 MW and an energy capacity of 400 MWh, providing a four-hour discharge duration.

For the CAPEX assessment, the unit cost per MWh includes the battery containers with integrated auxiliary systems (battery management system, electrical protections, thermal management, and fire protection), the power conditioning system (PCS) comprising inverters and associated transformers, and the required civil works and control infrastructure. The cost estimate also includes the on-site substation, consisting of the step-up transformer to the grid interconnection voltage and the associated switchgear and protection equipment. Grid connection costs are assumed to be included. Repowering, retrofitting, or reuse of existing assets are not considered.

Regarding operating expenditures (OPEX), the analysis considers a representative year of operation. Annual operation and maintenance costs are included, together with battery degradation effects, assuming an average capacity fade of approximately 1.5% per year. The cost associated with battery augmentation to compensate for this degradation is included in the annual O&M costs.

End-of-life costs, major reinvestments beyond battery augmentation, energy losses, and market-related revenues or costs are not considered. For the investment cost assessment of the BESS, the document [1] has also been used.

Table 17: Solution 2: investment and operational costs (BESS).

Item	**Value**	**Unit**
Installed capacity		
BESS power capacity	100	MW
BESS energy capacity	400	MWh
CAPEX (investment)		
Battery system investment	59.47	million €
Total investment	**59.47**	**million €**
OPEX (annual operational cost)		
BESS OPEX	3.41	million €/year
Total OPEX	**3.41**	**million €/year**

3.4 Solution 3: Dynamic reactive power compensation

3.4.1 Site selection and characteristics

Solution 3 focuses on the implementation of reactive power compensation systems to improve voltage profiles, reduce transmission losses, and enhance overall system efficiency. By installing reactive power support devices at strategically selected buses experiencing voltage violations, the solution addresses network performance issues without adding new generation capacity. The proposed compensation systems contribute to mitigating voltage drops, reducing reactive power flows across transmission lines, and improving voltage stability margins, thereby increasing the operational robustness and resilience of the power system.

As analysed previously in this document, there is a bus where voltage drops are significantly more severe compared to the rest of the network. This critical location corresponds to Bus 13 (Tarrega LV), which exhibits the most pronounced undervoltage conditions. In order to mitigate these voltage violations and improve the local voltage profile, a STATCOM is proposed to be installed at this bus. The reactive power support provided by the STATCOM enables dynamic voltage regulation at Tarrega LV, reducing voltage drops and contributing to an overall improvement in system performance and voltage stability.

3.4.2 Network Integration and load flow study

In order to analyse the impact of reactive power support on the voltage profile of the network, a STATCOM device was incorporated into the system model. The STATCOM was represented in *pandapower* as a controllable static generator (sgen) connected to bus 13. In this modelling approach the STATCOM is represented as a voltage-dependent reactive power source with zero active power injection.

The main electrical characteristics of the STATCOM were defined as follows:

- Connection bus: 13
- Rated apparent power: 60 MVA
- Reactive power limits: ±50 Mvar
- Voltage reference: 1.02 p.u.
- Voltage–reactive power gain: 10 Mvar/p.u.
- Voltage deadband: ±0.005 p.u.

The STATCOM operates by injecting or absorbing reactive power as a function of the voltage deviation at the point of connection with respect to the reference value. A relatively low control gain together with the introduction of a deadband was intentionally selected in order to ensure stable behaviour and to avoid excessive reactive power oscillations during the time-series simulations.

In order to analyse the impact of reactive power support on the voltage profile of the network, a STATCOM device was incorporated into the system model. The STATCOM was represented in *pandapower* as a controllable static generator (`sgen`) connected to bus 13. This modelling approach is widely adopted in steady-state and quasi-static time-series studies, where the STATCOM is represented as a voltage-dependent reactive power source with zero active power injection.

The STATCOM operates by injecting or absorbing reactive power as a function of the voltage deviation at the point of connection with respect to the reference value. A relatively low control gain together with the introduction of a deadband was intentionally selected in order to ensure stable behaviour and to avoid excessive reactive power oscillations during the time-series simulations.

Closed-loop voltage regulation using *pandapower* Controllers

To implement a closed-loop voltage regulation strategy within a time-series power flow framework, the *Controller* functionality provided by *pandapower* was employed. This approach enables the reactive power output of the STATCOM to be dynamically updated at each time step based on the resulting bus voltage, thus ensuring a consistent feedback control mechanism. Alternative approaches, such as manual simulation loops or iterative open-loop procedures based on precomputed voltage profiles, were also considered. Although these methods could potentially reproduce similar voltage regulation behaviour, they were not considered an ideal solution due to their limited scalability, reduced robustness, and poor integration with the existing time-series structure based on predefined load and generation profiles. The use of *pandapower Controllers* provides a clean and modular solution that naturally integrates with the `run_timeseries` framework. This guarantees that load variations, generation profiles, and voltage control actions are consistently applied within the same simulation loop.

Overall, this approach enables an effective quasi-static modelling of STATCOM voltage regulation, which is well suited for planning-oriented time-series studies where the objective is to evaluate voltage support performance rather than fast electromagnetic dynamics.

In Figure 28, it can be observed that the STATCOM provides a satisfactory voltage support performance. When the STATCOM is not activated, the minimum voltage magnitude reaches a value of 0.88 p.u. However, when the STATCOM is enabled, the minimum voltage increases to 0.944 p.u., representing a significant improvement in voltage regulation.

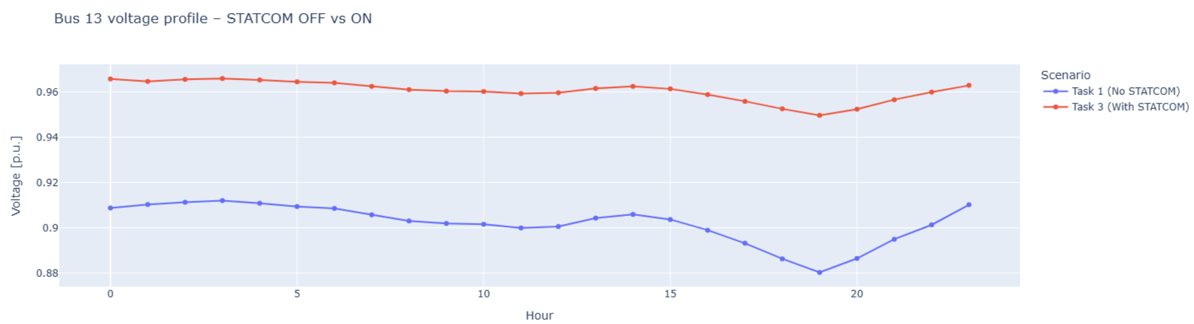

Figure 28: 24-hour voltage profile with STATCOM compensation K = 100

At this stage, it is reasonable to question whether the controller gain can be adjusted to further improve the system response, or whether the STATCOM device is over or under dimensioned in terms of rated power.

In order to assess the reactive power requirements, a logging mechanism was added within the STATCOM control class. Specifically, the reactive power calculated at each control step was recorded inside the control_step function. This approach allows direct observation of the reactive power injected by the STATCOM during the simulation. From these results, it can be concluded that the maximum injected reactive power is approximately 20 MVAr. Consequently, the apparent power rating of the STATCOM could be reduced, leading to lower investment costs while maintaining adequate voltage control performance.

Regarding the controller gain adjustment, an iterative analysis was carried out in order to evaluate whether higher gains lead to improved voltage regulation or, conversely, induce oscillatory or resonant behavior in the system. Although voltage magnitudes in per-unit provide a clear indication of steady-state improvement, they do not fully capture the dynamic behavior of the reactive power control. For this reason, the time evolution of the calculated reactive power was also analyzed.

An inflection point is observed around a gain value of $K = 400$, where the reactive power output begins to exhibit oscillatory behavior while the voltage improvement becomes marginal.

The results obtained for different controller gains can be seen in figure 29 and 30, also are detailed in Tables 21 and 22.

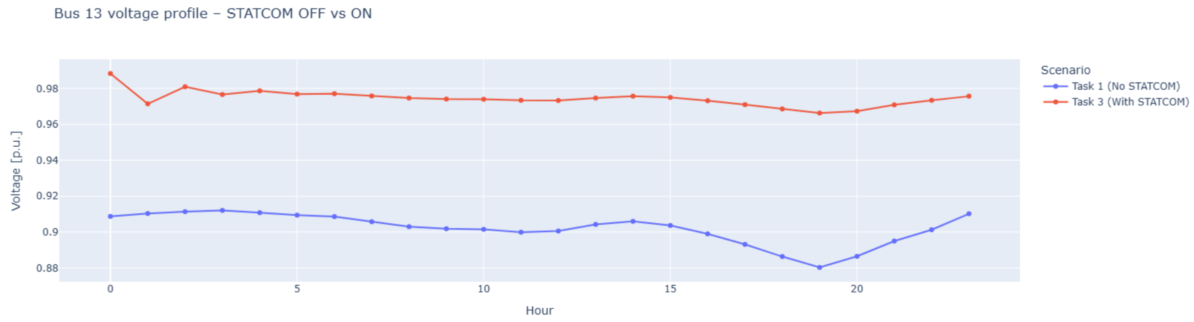

Figure 29: 24-hour voltage profile with STATCOM compensation K = 300

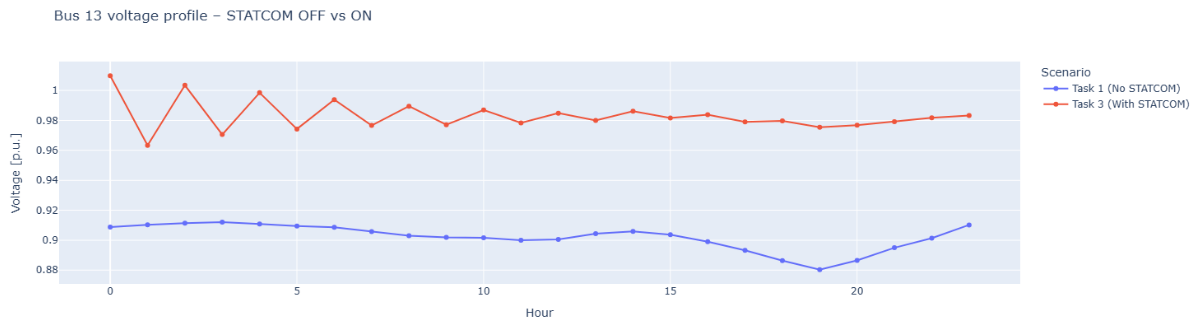

Figure 30: 24-hour voltage profile with STATCOM compensation K = 500

Overall, this study demonstrates that the integration of a STATCOM significantly improves the voltage quality of the network. Furthermore, it provides insight into the implementation of dynamic control processes within time-series power flow simulations and the analysis of quasi-dynamic behavior using *pandapower*. Finally, the required STATCOM rating was determined to be approximately 20 MVA, which represents an efficient compromise between performance and cost.

3.4.3 Investment and operating costs estimation

For the estimation of investment and operational costs, it is assumed that the STATCOM device is installed at a single network node. Since STATCOM units exhibit low energy consumption and relatively small operation and maintenance (O&M) requirements, operational costs are neglected in this study and approximated as zero for each installed unit.

The STATCOM technology selected for this analysis is based on ABB modular STATCOM units housed in NEMA 3R enclosures. Each unit provides a maximum continuous reactive power rating of approximately 1.9 MVAr and can be operated in parallel to achieve higher overall compensation levels. In order to reach a total reactive power capability of 20 MVAr, a total of ten STATCOM units are required.

Additionally, a power transformer is necessary to interface the STATCOM system with the transmission network, since the maximum operating voltage of the STATCOM units is 46 kV. For this purpose, a power transformer manufactured by Rockwill Electric Group [4] is considered. With a maximum rated capacity of up to 63 MVA depending on configuration, a single transformer is sufficient to meet the system requirements.

Based on these assumptions, the estimated investment costs for the STATCOM-based solution are summarized in Table 18.

Table 18: STATCOM solution: investment and operational costs

Item	**Value**	**Unit**
Installed capacity		
STATCOM reactive power capacity	20	MVAr
Number of STATCOM units	10	units
CAPEX (investment)		
ABB STATCOM unit cost (1.9 MVAr)	0.45	million €/unit
Total STATCOM units investment	4.50	million €
Power transformer (110/66 kV, 63 MVA)	0.20	million €
Total investment	**4.70**	**million €**
OPEX (annual operational cost)		
STATCOM OPEX	0.00	million €/year
Total OPEX	**0.00**	**million €/year**

3.5 Coparation of the three solutions

This section presents a comparative economic assessment of the three mitigation strategies analyzed in this work:

(i) the integration of additional renewable energy sources (RES),

(ii) the deployment of a battery energy storage system (BESS) at a critical node,

(iii) the installation of a STATCOM for voltage support.

Although these solutions address different aspects of network performance and are not technically equivalent, a comparative economic analysis provides valuable insight into their relative cost implications and practical feasibility.

From a purely investment-based perspective, the STATCOM solution represents by far the lowest capital expenditure, with a total estimated investment of 4.7 million €, as summarized in Table 18. This solution focuses exclusively on reactive power compensation and voltage regulation, without contributing active power or energy to the system. Operational costs associated with the STATCOM are considered negligible due to its low losses and limited maintenance requirements.

In contrast, the BESS-based solution entails the highest capital and operational costs among the analyzed options. As shown in Table 17, the battery system requires an investment of approximately 59.5 million €, together with an annual operational cost of 3.41 million €. Despite its high cost, the BESS provides a wide range of system services, including energy shifting, peak shaving, frequency support, and voltage regulation, which cannot be achieved by the STATCOM alone. Therefore, its economic assessment must be interpreted in light of its multi-service capability rather than voltage control only.

The RES-based solution, detailed in Table 16, presents an intermediate investment level of 46.5 million €, with relatively low annual operational costs compared to the BESS solution. This option increases the total generation capacity of the system by incorporating new solar and wind installations, as well as the necessary grid reinforcements. While this strategy contributes to long-term decarbonization and active power supply, its effectiveness in mitigating local voltage issues is indirect and strongly dependent on generation availability and operating conditions.

Overall, the comparison highlights that the STATCOM solution offers the most cost-effective approach for addressing voltage regulation issues when the problem is primarily related to reactive power management. The BESS solution, although significantly more expensive, provides broader operational flexibility and additional system services that may justify its higher cost in more complex scenarios. Finally, the RES expansion strategy represents a long-term investment focused on generation capacity and sustainability rather than targeted voltage control. Consequently, the selection of the most suitable solution depends not only on economic considerations, but also on the specific operational objectives and planning horizon of the network.

4 Task IV Use Case Design and SGAM

The purpose is to design a use case and map it to the Smart Grid Architecture Model (SGAM). This task demonstrates how system improvements from previous sections can be applied in a practical scenario. The scope includes use case definition, SGAM layer mapping, and traceability to previous tasks as specified in the course requirements.

4.1 Use Case

The potential use cases for the given scenario are:

- **Critical Load Supply Assurance:** Ensures continuous power supply to critical healthcare facilities during network contingencies by leveraging Phase 1 network redundancy and local generation resources.
- **Renewable Energy Integration and Grid Support:** Manages the integration of renewable energy sources (solar and wind) to reduce import dependence while maintaining voltage stability and system reliability.
- **Voltage Regulation and Power Quality:** Maintains voltage profiles within acceptable limits across all buses, particularly during peak demand periods, using reactive power compensation and distributed generation.
- **Demand Response and Peak Shaving:** Reduces peak-period imports and operating costs by coordinating local generation, storage systems, and demand-side management for critical facilities.
- **N-1 Contingency Management:** Automatically detects and responds to single-element outages, ensuring supply continuity through alternative power flow paths enabled by Phase 1 network reinforcement.

4.2 Use Case: Critical Load Supply Assurance for the Healthcare District in Tàrrega

Tàrrega is building a new health and wellness district with hospitals, clinics, and elderly care facilities. Ensuring uninterrupted, clean energy is critical. The project envisions a smart grid with combined heat and power (CHP), storage, and predictive maintenance to prioritise critical loads.

Table 19: SGAM domains and roles adapted to the Tàrrega healthcare smart grid scenario

Domain	**Actors or roles in the scenario**
Customer	Healthcare facilities in Tàrregas new health and wellness district (hospitals, clinics, elderly care centers)
Markets	Electricity market operators and regulatory framework governing energy prices and imports
Service Provider	Energy service and ICT providers supporting monitoring, predictive maintenance, and optimization
Operations	System operators responsible for real-time grid operation and contingency management
(Bulk) Generation	External grid supply and centralized generation supporting the local network
Transmission	Transmission System Operator (TSO) managing high-voltage power transport
Distribution	Distribution System Operator (DSO) operating the local MV/LV network, CHP, RES, and storage

Actors

Preconditions

- Phase 1 network reinforcement is operational, providing meshed topology and N–1 redundancy.
- Renewable energy sources (solar and wind) are integrated and operational.
- Monitoring, protection, and control systems are functional.
- Healthcare facilities are connected to the distribution network.
- Backup power systems at healthcare facilities are available and tested.

Triggers

- Detection of a single-element outage (transmission line or transformer).
- Voltage deviation outside acceptable limits (below 0.9 p.u. or above 1.1 p.u.).
- Reduction of import capacity or external grid contingency.
- Scheduled maintenance requiring network reconfiguration.
- Emergency situation requiring priority supply to critical healthcare loads.

Main Flow

1. System monitoring detects a contingency event such as a line outage, transformer failure, or voltage deviation.
2. Protection systems automatically isolate the faulted element while maintaining system stability.
3. The network control system evaluates available power flow paths using the Phase 1 meshed topology.
4. Local renewable generation is dispatched to compensate for reduced import capacity.

5. Voltage control systems activate reactive power compensation if voltage deviations are detected.
6. Load prioritization mechanisms ensure uninterrupted supply to critical healthcare loads.
7. System monitoring continuously evaluates voltage profiles, line loadings, and the supply–demand balance.
8. If local generation is insufficient, on-site backup power systems at healthcare facilities are activated.
9. Network reconfiguration is executed to optimize power flows and minimize losses.
10. The system returns to normal operation once the contingency is resolved or alternative supply paths are established.

Alternative Flows

- **Alternative Flow A – Insufficient Local Generation:** If renewable generation and CHP units cannot meet demand, healthcare facility backup systems are activated. Non-critical loads may be curtailed, and additional import capacity is requested if available.
- **Alternative Flow B – Multiple Contingencies:** If multiple elements fail simultaneously, emergency procedures are activated. Critical loads are supplied through dedicated paths where possible, and the system operates in a degraded mode until repairs are completed.
- **Alternative Flow C – Voltage Instability:** If voltage deviations persist, additional reactive power resources are activated, generation dispatch is adjusted, and non-critical load shedding may be applied.
- **Alternative Flow D – Communication Failure:** In case of communication failure, the system operates using local autonomous control functions with pre-configured protection settings. Manual intervention may be required for complex scenarios.

Postconditions

- Critical healthcare loads maintain continuous power supply throughout the contingency.
- System voltage profiles remain within acceptable limits (0.9–1.1 p.u.).
- Network components operate within thermal limits (line loading below 80%).
- System monitoring confirms stable or degraded but secure operation.
- Contingency events are logged for analysis and future system improvement.

4.3 SGAM Mapping for the use case

SGAM (Smart Grid Architecture Model) is used in this work as a traceability framework that links the planning solutions developed in previous tasks (network reinforcement, RES integration, storage, and reactive power compensation) to an operationally meaningful use case. To keep the architectural description compact and aligned with the scope of this course project, the SGAM analysis focuses on the Layer dimension (Business, Function, Information, Communication and Component) and relates each layer to the Domain–Zone view when relevant (Distribution / DER / Customer Premises across Process to Market zones).

4.3.1 Business Layer

The Business layer captures the operational objectives, stakeholders, and business processes that justify the use case.

- **Business objectives**
  - Guarantee continuity of supply for critical loads under N–1 contingencies.
  - Maintain power quality (voltage within contractual limits) during normal operation and post-contingency states.
  - Reduce reliance demonstrates improved resilience through local flexibility (RES, storage and reactive support).
- **Stakeholders / roles**
  - TSO/Network Operator: system security and contingency coordination.
  - DSO: local service continuity, voltage quality and customer supply management.
  - Healthcare Facility Operator: critical load prioritisation and on-site backup operation.
  - DER Operators: availability and dispatch of local generation/storage.
- **Regulatory / operational constraints**
  - N–1 security criterion for credible single-element outages.
  - Voltage quality requirements at customer connection points.
  - Thermal loading limits on transmission/distribution assets.

4.3.2 Function Layer

The Function layer describes the logical functions that must be performed to achieve the use case objectives.

- **Detection**
  - Grid state monitoring (bus voltages, line loadings, power balance).
  - Contingency detection (line/transformer outage, abnormal voltage profile).
  - Critical load status monitoring (availability, priority class).
- **Decision and control**
  - Security assessment and N–1 verification (post-contingency feasibility check).
  - Corrective actions selection: network reconfiguration, voltage support activation, DER setpoint updates.
  - Critical load prioritisation logic.
- **Actuation**
  - Switching actions (reconfiguration to alternative supply paths enabled by Phase 1 meshing).
  - Voltage control actions (reactive power support / STATCOM setpoints, tap changer actions if applicable).

  - Active power actions (DER dispatch and/or BESS charge/discharge schedule adjustment).
  - Backup activation at customer premises (only if grid-side actions cannot maintain supply).

4.3.3 Information Layer

The Information layer defines the key data objects exchanged between actors and functions. In this use case, information exchange is structured around operational measurements, setpoints, and event logs.

- **Measurements**
  - Voltage magnitudes at critical buses.
  - Line loading values and breaker/switch status.
  - Active/reactive power injections from RES/BESS/STATCOM.
  - Critical load demand and priority classification (critical vs non-critical).
- **Commands and setpoints**
  - STATCOM reactive power setpoint / voltage reference commands.
  - BESS active power setpoint (charge/discharge) and state-of-charge limits.
  - DER curtailment / dispatch instructions.
  - Switching commands for network reconfiguration.
- **Events and logs**
  - Fault and protection events (trip signals, outage identification).
  - Post-contingency performance reports.

4.3.4 Communication Layer

The Communication layer specifies how information is transported between field devices, substations, control centres, and customer premises.

- **Field/Station to Operation center**
  - Substation measurements and status are transmitted to SCADA and EMS.
  - Typical characteristics: periodic telemetry + real time event-based alarms.
- **Operation to Field/Station**
  - Dispatch commands for STATCOM and BESS setpoints.
  - Switching commands for reconfiguration (where remotely controlled).
- **Operation between Customer premises for critical load coordination**
  - Exchange of load priority state, backup availability and emergency operating modes.

4.3.5 Component Layer

The Component layer maps the functions to physical equipment and systems. Components are grouped according to the main SGAM domains (Distribution, DER, Customer Premises).

- **Grid infrastructure**
  - Phase 1 reinforced transmission network (meshed topology and additional interconnections).
  - Substations, circuit breakers, disconnectors, protection relays.
  - Measurement devices for feeding SCADA and EMS.
  - Control centre systems for managing SCADA and EMS.
- **DER domain (flexibility assets)**
  - Renewable generation plants with inverters and plant controllers.
  - BESS with power conversion system, SoC management logic and controllable setpoints.
  - STATCOM at the critical bus for reactive compensation device for voltage regulation.
- **Critical loads**
  - Critical/non-critical separation logic for each load.
  - On-site backup systems.
  - Smart metering / monitoring devices providing demand and status signals.

5 Conclusions

The 110 kV network model was successfully implemented in PandaPower and the 24-hour time-series load flow converged for all hours. The main technical issue is voltage performance: several buses fall below 0.9 p.u. during peak demand, while line loading stays clearly below the 80% limit, so thermal overload is not the bottleneck.

From the economic side, the system still needs the Terrassa interconnection during peak hours, even if it exports energy in other periods. The annual failure-cost estimation is mainly driven by line failures, suggesting that reinforcing the most critical corridors and improving voltage support should be prioritised in the next planning stage.

The network serving Lleida, Tàrrega, Manresa, and Montblanc suffers from voltage violations ($V_{\min} = 0.859$ p.u.), N-1 non-compliance, and peak import dependence (23.7 MW). In Phase 1, we reinforced through 194 km of targeted double-circuit lines which cost 48.5 M€investment. The reinforcement successfully eliminated all voltage deviations, achieves full N-1 compliance for critical contingencies, and reduced maximum line loading to 32.89% which is well below 80% limit. The only minor remaining issue is transmission losses at 2.29% , slightly above 2% target, equivalent to 42 MWh/year excess. Thus, in Phase 2, we would address residual import dependence and decarbonisation through RES integration, BESS deployment, or STATCOM compensation.

Phase 2 has addressed the need for reinforcing the electrical network by evaluating different upgrade strategies aimed at mitigating operational constraints and improving overall system performance. Three alternative solutions have been analyzed: the installation of a STATCOM, the integration of additional renewable energy sources (RES), and the deployment of a battery energy storage system (BESS). Each option has been assessed from both a technical and an economic perspective.

The analysis shows that the STATCOM solution represents a technically effective and economically efficient approach when the main limitation of the network is related to reactive power management and voltage regulation. Its relatively low investment and operational costs make it a suitable solution for addressing localized voltage issues without significantly altering the active power balance of the system.

The RES-based solution contributes to increasing local generation capacity and supports long-term decarbonization objectives. However, its effectiveness is strongly dependent on generation availability and operating conditions, and it may require additional grid reinforcements to accommodate higher penetration levels. As a result, this option is better suited for medium to long term planning strategies rather than for addressing immediate operational constraints.

The BESS based solution, provides the highest level of operational flexibility among the analyzed options. By enabling peak shaving, import reduction, and bidirectional power exchange, this solution can address multiple system needs simultaneously. Nevertheless, it entails significantly higher capital and operational costs, and its assessment must therefore be justified by its several services capability rather than by a single technical objective.

Overall, the conclusions from Phase 2 highlight that there is no universally optimal solution for network reinforcement. The selection of an appropriate upgrade strategy strongly depends on the specific nature of the network constraints, the desired operational objectives, and the planning horizon.

6 Annexes

A N-1 contingency analysis results for line outages

Table 20: N-1 contingency analysis results for line outages

Line Fail	Simulation	N-1 Pass	Violated Buses	Violated Lines
0	True	False	[9]	[]
1	True	False	[9]	[]
2	True	False	[9]	[]
3	True	False	[9]	[]
4	True	False	[9]	[]
5	True	False	[9]	[]
6	True	False	[9]	[]
7	True	False	[9]	[]
8	True	False	[9]	[]
9	True	False	[9]	[]
10	True	False	[7, 9]	[]
11	True	False	[9]	[]
12	True	False	[9]	[]
13	True	False	[8, 9]	[]
14	True	False	[9]	[]

B Output results in Phase 2 solution 3

Table 21: Output STATCOM K = 300

Simulation progress [%]	V_m **[p.u.]**	$q_{\mathbf{new}}$
0	0.9541	19.7597
8	0.9893	9.1996
12	0.9721	14.3761
17	0.9814	11.5773
21	0.9758	13.2454
25	0.9778	12.6577
29	0.9763	13.1179
33	0.9753	13.4147
38	0.9741	13.7702
42	0.9740	13.7934
46	0.9738	13.8503
50	0.9729	14.1262
54	0.9738	13.8677
58	0.9756	13.3064
62	0.9757	13.2961
67	0.9743	13.7235
71	0.9721	14.3686
75	0.9698	15.0574
79	0.9672	15.8538
83	0.9653	16.4196
88	0.9696	15.1183
92	0.9722	14.3522
96	0.9746	13.6144
100	0.9767	12.9855

Table 22: Output STATCOM K = 500

Simulation progress [%]	V_m **[p.u.]**	$q_{\mathbf{new}}$
0	0.9541	32.9328
8	1.0108	4.5947
12	0.9641	27.9692
17	1.0039	8.0486
21	0.9698	25.1171
25	0.9976	11.1758
29	0.9737	23.1344
33	0.9922	13.9199
38	0.9750	22.5166
42	0.9888	15.5867
46	0.9769	21.5468
50	0.9860	17.0036
54	0.9787	20.6514
58	0.9871	16.4355
62	0.9810	19.4947
67	0.9848	17.6055
71	0.9788	20.6212
75	0.9805	19.7353
79	0.9752	22.3790
83	0.9766	21.7216
88	0.9787	20.6423
92	0.9816	19.2014
96	0.9829	18.5638
100	0.9851	17.4469

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